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A comprehensive deep learning framework for rice variety classification with real-time deployment

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ABSTRACT

Rice is a staple food for more than half of the global population and a cornerstone of agricultural economies. With the growing digitalization of agri-food supply chains, automated rice variety recognition has become essential for breeding, seed authentication, quality assurance, and fraud prevention. However, most existing studies are constrained by small datasets often including fewer than twenty varieties limiting model generalization and real-world applicability. To address this gap, we developed a comprehensive deep learning framework that evaluates (i) transfer learning models, (ii) a stacked ensemble of 13 meta-classifiers, and (iii) both raw and pretrained Vision Transformer (ViT-B/16) models on a large-scale and diverse image dataset comprising 62 rice varieties. Experimental results demonstrate that the stacked ensemble achieved the highest classification accuracy, outperforming individual CNN and transformer architectures. For real-world use, three top-performing models VGG16, the stacked ensemble, and Google's ViT-B/16 are deployed through a lightweight web platform (RiceVision) and a mobile application, enabling real-time rice variety identification. This study underscores the translational potential of AI-driven computer vision in advancing smart agriculture and digital seed supply chain management.

1. Introduction

Rice (*Oryza sativa* L.) is one of the most important staple crops worldwide [1], providing the primary source of calories for more than half of the global population [2]. In rice-dependent countries such as Bangladesh, India, China, and other Asian nations [3], it plays a vital role in food security, rural livelihoods, and economic stability [4]. Beyond its nutritional value, rice cultivation employs millions of farmers and contributes substantially to agricultural GDP and global trade [5]. Agriculture, forestry, and fishing generate nearly 4 trillion USD annually [6], with crops accounting for 2.8 trillion USD [7]. Cereals account for ~30 % of this value (≈840 billion USD) [8], and rice alone contributes around 315 billion USD, which is ~11 % of the global crop sector [9]. Projections further suggest that the rice market will expand to 358.6 billion USD by 2030 [10], underscoring its strategic role not only as a dietary staple but also as a driver of economic growth and international trade.

Despite its importance, rice variety classification remains a complex challenge [11]. Thousands of cultivars exist worldwide, differing subtly in grain size, shape, colour, texture, and chemical composition [12]. Accurate recognition depends on both surface textures and internal geometric features. Rice kernels may vary from smooth and transparent to rough, porous, or chalky, traits often linked to processing quality. Morphological descriptors such as kernel length, width, convex hull, curvature, and head shape provide additional structural cues [13]. While these features have long been used in manual inspection, such methods are subjective, time-consuming, and unsuitable for large-scale industrial or research applications. These difficulties highlight the pressing need for automated, reproducible, and scalable rice classification systems that can support breeding programs, consumer protection, and global market transparency.

Recent advances in Artificial Intelligence (AI), particularly in Machine Learning (ML) and Deep Learning (DL), have demonstrated remarkable potential in crop and grain classification tasks [14].

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Convolutional Neural Networks (CNNs) [15,16], support vector machines (SVMs) [17], and, more recently, Vision Transformers (ViTs) [18] have achieved strong performance in controlled experiments. However, the effectiveness and generalizability of these models are fundamentally constrained by the size, diversity, and quality of the training data. Most existing rice variety classification studies [19–50] consider fewer than ten varieties and rely on datasets containing only a few hundred to a few thousand images, often captured under narrow and non-standardized imaging conditions.

A critical gap in the literature is the absence of a unified, large-scale rice benchmark that consolidates multi-source datasets under consistent imaging and labelling protocols. No prior work has systematically integrated heterogeneous rice datasets into a harmonized standard or jointly evaluated CNNs, stacked ensemble meta-classifiers, and transformer architectures within a single experimental framework. Furthermore, despite the growing interest in AI for agricultural applications, existing studies rarely address real-time deployment challenges, such as inference speed, device compatibility, and robustness under natural field conditions. This lack of standardized data and deployment-focused evaluation presents a significant barrier to developing scalable, reliable, and practically deployable AI-driven rice classification systems.

To address these limitations, this study combined the first harmonized 62-variety rice benchmark, constructed by integrating three independently verified data sources and standardizing all samples through a controlled imaging pipeline. Unlike previous works limited by small class sizes or single-model investigations, our framework systematically evaluates deep feature extraction, ensemble fusion strategies, and transformer-based attention mechanisms under a unified benchmark. The resulting Rice-62 dataset comprises 37,200 training and 9300 testing images and includes 20 field-collected varieties authenticated by experts, 5 international varieties from public repositories, and 37 varieties contributed by agricultural research teams. All samples are collected and curated under consistent imaging conditions to preserve fine-grained morphological variations essential for classification. Leveraging this dataset, we develop a comprehensive four-fold evaluation pipeline: (i) transfer learning with four CNN backbones to benchmark deep feature extraction, (ii) ensemble learning with 13 meta-classifiers to improve robustness, (iii) a raw ViT model to establish transformer baselines, and a pretrained Google ViT-B/16 model to assess state-of-the-art transformer performance, and (iv) Finally, we demonstrate translational potential by deploying the proposed ensemble model as a web-based system and mobile app, enabling real-time rice variety recognition for breeders, regulators, and industry stakeholders.

The contributions of this work are summarized as follows:

i. We conduct a rigorous evaluation across multiple deep learning paradigms including four CNN backbones, stacked-ensemble method using meta-classifier, and two transformer models on a diverse, large-scale image dataset encompassing 62 rice varieties, thereby establishing the most comprehensive performance benchmark for rice variety recognition to date.
ii. Beyond experimental evaluation, we deploy the three best-performing models VGG16 from transfer learning, the stacked ensemble framework, and Google's pretrained ViT-B/16 within web-based and only VGG16 for lightweight mobile applications. This integrated deployment enables real-time rice variety classification and demonstrates the practical and translational potential of AI-driven computer vision in agricultural informatics.
iii. We establish a reproducible and leakage-free experimental protocol using stratified splitting, 5-fold cross-validation, and a complete robustness assessment including confusion matrices, ROC curves, and misclassification analysis.
iv. We analyse the computational cost versus accuracy across all evaluated models, revealing the stacked ensemble as the most efficient and accurate architecture for real-time agricultural use.

2. Related work

Research on rice variety recognition has evolved steadily from traditional image analysis to modern ML and DL approaches. Early studies largely depended on handcrafted morphological and texture descriptors, later enhanced with classical ML classifiers, while recent work has increasingly adopted convolutional networks, lightweight models, and ensemble strategies capable of high accuracy and near real-time deployment.

The first automated attempts can be traced back to the mid-1990s, when two-dimensional imagery was applied to four rice types [19]. In 2002, Wee et al. [20] introduced a fuzzy ART-map neural network that achieved 95.4 % accuracy for rice quality classification, while Marini et al. [21] applied a counter-propagation ANN to predict 11 varieties with close to 90 % accuracy. Hobson et al. [22] advanced the field in 2007 by employing digital image analysis for eight varieties, and Guzman and Peralta [23] developed an ANN-based classifier that reached 97 % accuracy across five Philippine rice groups. Rad et al. [24] used grain color and texture features to achieve 96.7 % accuracy on five classes, while Nagendra et al. [25] tested a chemical-based method, reporting 85 % accuracy.

In the following decade, researchers broadened the use of classical ML. Huang et al. [26] applied SVMs to 79 varieties, reporting 79.7 % accuracy, while Aki et al. [27] tested the Non-Nested Generalization Algorithm, achieving 90.5 %. Sumaryanti et al. [28] demonstrated the effectiveness of learning vector quantization with 96.6 % accuracy on 10 types, and Zareiforoush et al. [31] reported 98.7 % using ANN on a smaller dataset. Kuo et al. [30] tested sparse representation classifiers across 30 varieties with 89.1 % accuracy, and hybrid pipelines such as backpropagation neural networks with wavelet decomposition were also explored to improve feature representation [29].

The adoption of DL after 2017 marked a turning point. Lin et al. [32] employed a deep CNN, reaching 95.5 % accuracy on three varieties, while Rexce et al. [33] used an MLP on 13 types, obtaining 92.3 %. Wah et al. [34] combined segmentation with KNN for three varieties, and Cinar and Koklu [35] compared different AI methods, achieving 93.0 %. Hue et al. [36] applied morphological feature extraction with 95.5 % accuracy, and Dheer and Singh [37] achieved 99.2 % across eight varieties using ML classifiers. Since 2020, the field has expanded rapidly with larger datasets, hybrid approaches, and lightweight architectures. Ibrahim et al. [38] conducted a comparative study of ANN models with 93.3 % accuracy, while Weng et al. [39] combined hyperspectral imaging with DL to achieve 98.6 %. Saxena et al. [40] reported 99.9 % accuracy using classical ML for five varieties, and CNNs were further tested by Lee and Tay [41] on seven types. Nair et al. [42] designed a fuzzy genetic cascading system, achieving 94 %, while Jeyaraj et al. [43] developed a DL-based pipeline for real-time recognition of five varieties. Lightweight models have become increasingly attractive: Tasci et al. [44] implemented a quantized LeNet-5, achieving 99.9 %, and Iqbal et al. [45] employed MobileNetV2, reaching 99.7 % accuracy across five varieties. Kang et al. [46] integrated fluorescence hyperspectral imaging with backpropagation NNs and Random Forests, reporting 95.3 %, while ensemble SVMs classified three varieties with 96 % accuracy [47]. More recently, Verma et al. [48] applied EfficientNet-B1 on a five-class dataset, obtaining 99.9 %, though limited by dataset diversity. Larger studies include Islam et al. [49], who achieved 100 % accuracy on 20 varieties using ensemble learning, and their subsequent work [50], which applied ConvNeXt on 27,000 images across 20 varieties, reaching 99.6 %. To provide a clearer comparative view, Table 1 summarizes representative studies on rice variety classification, highlighting the datasets, methods, and reported performance.

Overall, the literature charts steady progress from handcrafted features and classical ML to CNNs, lightweight networks, and ensembles. Although, recent works have also explored transformer-based and hybrid deep architectures for agricultural tasks, including the QRBiLSTM-MHSA network [67] for rice yield prediction and

Table 1
Summary of Classification methods and results for different dataset sizes and varieties.

Study	Year	Number of Samples	Classes	Publicly Accessible	Methodology	Result Types	Results	Real-Word Application
S1 [19]	1996	400	4	×	Dimension and Shape Factor	Probability	68.3 % −95.4 %	×
S2 [61]	2002	NA	2	×	Fuzzy Art-map Neural Network	Accuracy	95.40 %	×
S3 [21]	2004	1779	11	×	Counter-Propagation Artificial Neural Network (CP-ANN)	Accuracy	91 %−99 %	×
S4 [22]	2007	400	8	×	Unsupervised Clustering Techniques	Dendrogram	Not Specified	×
S5 [23]	2008	5720	52	×	Multi-Layer Perceptron	Accuracy	66.67 %−100 %	×
S6 [52]	2011	500	5	×	Segmentation And Back-Propagation Neural Network-Based Classifier	Accuracy	96.67 %	×
S7 [53]	2012	2100	7	×	Imperialist competition algorithm and SVM	F-Measure	90.02 %−95.14 %	×
S8 [54]	2012	1500	5	×	Segmentation and a Back-propagation neural network-based classifier	Accuracy	98.40 %	×
S9 [55]	2013	Not Specified	1	×	Multi-Class SVM	Accuracy	>86 %	×
S10 [56]	2014	477	3	×	Principal Component Analysis	Accuracy	75 %−80 %	×
S11 [57]	2021	10800	6	×	Logistic Model Tree, Regression, J48 Tree, Meta Bagging, Meta Attribute Selected Classifier	Accuracy	97.4 %	×
S12 [29]	2016	1200	4	×	BPNN, SVM, KNN, Naive Bayes	Accuracy	96.25 %−100 %	×
S13 [32]	2017	7399	3	×	DCNN	Accuracy	95.50 %	×
S14 [58]	2017	Not Specified	5	×	Scattered and Heap Fashion Image Processing	Scatter Plot	Various Results	×
S15 [34]	2018	Not Specified	1	×	KNN	Accuracy	83 %−100 %	×
S16 [35]	2019	3810	2	×	LR, MLP, SVM, DT, RF, NB and k-NN	Accuracy	93.02 %	×
S17 [38]	2020	90	3	×	SVM and ANN	Accuracy	93.34 %	×
S18 [39]	2020	4320	10	×	principal component analysis network	Accuracy	98.57 %	×
S19 [59]	2021	75,000	5	✔	ANN, DNN, CNN	Accuracy	99.87 %−100 %	×
S20 [60]	2021	75,000	5	✔	Not Specified	Not Specified	Not Specified	×
S21 [42]	2022	3810	2	×	Genetic Fuzzy Cascading System	Accuracy	94.36 %	×
S22 [43]	2022	857	9	×	CNN	Accuracy	96.40 %	✔
S23 [41]	2022	700	7	×	ANN and CNN	Accuracy	88.3 %−99.5 %	×
S24 [40]	2022	75,000	5	✔	Random Forest	Accuracy	99.85 %	×
S25 [45]	2023	75,000	5	✔	CNN	Accuracy	99.20 %	×
S26 [44]	2023	75,000	5	✔	Quantized Neural Network	Accuracy	99.87 %	×
S27 [47]	2024	45,000	3	✔	Ensemble Learning	Accuracy	96.74 %	×
S28 [46]	2024	550	5	×	Principal Component Analysis, BP Neural Network, Random Forest	Accuracy	95.30 %	×
S29 [48]	2025	75,000	5	✔	EfficientNet-B1	Accuracy	99.87 %	×
S30 [49]	2025	27,000	20	✔	Ensemble Model	Accuracy	100 %	✔
S31 [50]	2025	27,000	20	✔	Next Generation Convolutional Neural Network	Accuracy	99.60 %	×

iMobileTransformer [68] for lightweight disease recognition. Incorporating these transformer-driven approaches into grain-level classification remains unexplored, highlighting the need for our comparative analysis between CNN backbones, stacked ensembles, and two variants of Vision Transformers. In addition, most studies still employ small datasets (often <10 to 20 varieties), provide limited cross-architecture comparisons, and rarely evaluate transformer-based models despite their success in general vision. As emphasized by Islam et al. [51], standardized, large-scale benchmarks are essential for robust, generalizable models. Motivated by these limitations, our study combines three different sources' datasets and introduces a 62-variety dataset to conduct a comprehensive evaluation spanning CNN transfer learning, ensemble meta-learning, and transformer models (raw ViT and pre-trained Google ViT-Base/16), followed by web deployment of a lightweight model for real-time use. In doing so, the study not only provides a benchmark for future research but also demonstrates the translational potential of AI in real-world rice classification.

3. Methodology

3.1. Dataset description

In data-driven classification, whether using traditional ML or DL, the quality, diversity, and representativeness of the dataset are decisive for model accuracy and generalizability. A carefully curated dataset enables algorithms to capture meaningful discriminative features, reduces bias and overfitting, and supports robust deployment in real-world settings. This is especially crucial in agricultural applications such as rice variety recognition, where natural variability in grain appearance adds substantial complexity. As shown in Fig. 1, rice grains exhibit multiple visual characteristics such as smoothness, roughness, porosity, chalkiness, transparency, and color, alongside semi-rough surface types. In addition to these qualitative attributes, morphological features, including grain length, width, and contour curvature, provide crucial quantitative descriptors. Together, these visual and structural features form the foundation of the dataset, ensuring that the trained models can robustly learn discriminative patterns for accurate classification across different rice varieties.

To address this need, we compiled a benchmark dataset by combining three different sources: Dataset-01 (Aruzz22.5 K [64]), our self-curated collection; Dataset-02 [35] from Cinar and Koklu; and Dataset-03 [63] from Bangladeshi researchers in collaboration with BRRI. All images are standardized to 224 × 224 pixels, converted to JPG format, and balanced with uniform class distribution. Each sample depicts a single grain against a black background at ~5× magnification, ensuring clear visibility of texture and morphology. An 80/20 train-test split has been applied, and data augmentation has been used to improve robustness. To enhance model generalization and maintain balanced representation across all 62 rice varieties, data augmentation is applied uniformly to the training set. Augmentation techniques including rotation, horizontal flipping, and brightness adjustment are designed to preserve the natural morphology of rice kernels while introducing beneficial variation in illumination and orientation. This process increased the number of training samples per class to an equal level, preventing class imbalance and reducing the risk of overfitting. Importantly, no augmented samples are included in the test set, ensuring that evaluation remained unbiased and fully representative of real, non-synthetic images. Overall, the augmentation strategy strengthened the dataset's diversity without altering the underlying visual characteristics essential for accurate fine-grained classification. The final dataset comprises 37,200 training and 9300 test images across 62 varieties, representing the most comprehensive benchmark to date for rice classification. Detailed statistics for each variety are provided in Table 2. Another Table A1 in Appendix A represents visual features comparison summary of all 62 types rice.

3.1.1. Rice variety authentication

All field-collected rice varieties are authenticated by expert agronomists from different places. Dataset-01 [62] is validated by domain expert from Bangladesh, Dataset-02 [35] is internationally well-recognized rice variety dataset, and Dataset-03 [63] is validated by the domain expert from BRRI organization. Dataset from Bangladesh (Dataset-01 and Dataset-03) are authenticated by BRRI's official morphological descriptor guidelines including grain length–width ratio, chalkiness, surface glossiness, and husk curvature. Any ambiguous samples are re-evaluated through a two-step manual inspection. Each of the 62 rice varieties included in this study is an officially recognized and independently documented cultivar with distinct morphological and agronomic descriptors. As a result, there is no possibility of class-level duplication across the three merged datasets, since every variety represents a unique and non-overlapping category. Thus, while we standardized imaging conditions for consistency, varietal duplication is not a concern. To prevent information leakage, we applied a strict stratified 80/20 split ensuring that no grain image from the same physical sample appeared in both training and testing sets. In addition, a 5-fold cross-validation protocol has implemented, and all experiments are repeated five times to compute mean accuracy and variance. This multi-run evaluation provides a statistically reliable measure of generalization and eliminates the risk of inflated performance.

3.2. Model architectures

To systematically evaluate rice variety classification, we designed a multi-stage architecture pipeline. First, we fine-tuned and optimized transfer learning–based CNN backbones, including VGG16, MobileNetV2, InceptionV3, and Xception, which has been selected for their proven strengths in feature extraction and classification. Building on these results, we employed a stacked ensemble strategy that integrated feature representations from the best-performing CNNs and evaluated them using 13 well-established meta-classifiers to maximize predictive performance. We then investigated transformer-based approaches, applying both a raw ViT and the pretrained Google ViT-B/16 with fine-tuning to assess their capacity to model complex patterns in the dataset. Finally, to ensure practical usability, the most efficient model has been deployed as a lightweight web-based system. The details of this architecture are presented in the following subsections.

3.2.1. Transfer-learning models

Transfer learning is a widely adopted strategy in image classification, allowing models to leverage pretrained weights from large-scale datasets such as ImageNet and adapt them to domain-specific tasks. By initializing models with pretrained weights, low-level features such as edges, shapes, and textures are already well learned, allowing the training process to focus on higher-level, task-specific features. In this

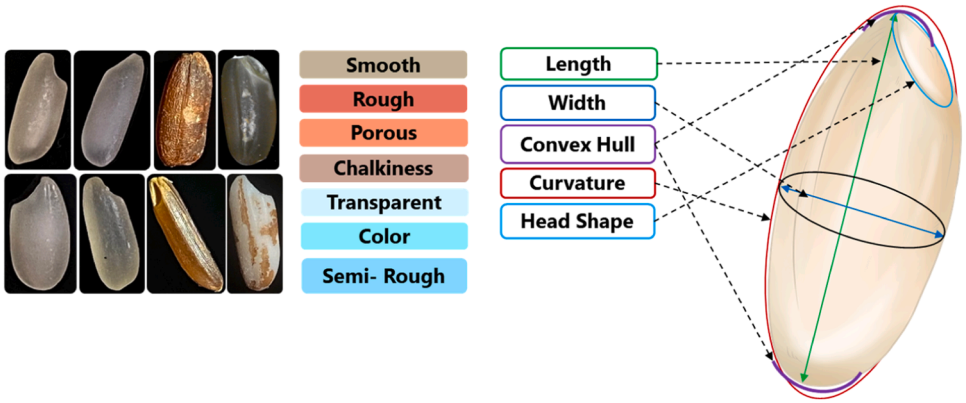


Fig. 1. Morphological and surface-based features used in rice grain characterization.

Table 2
Dataset description.

Individual Datasets	Sl. No.	Variety	Original Images	Augmented Images	
				Train (80 %)	Test (20 %)
Dataset-01 [62]	1	Subol Lota	232	600	150
	2	Bashmoti	230	600	150
	3	Ganjiya	233	600	150
	4	Shampakatari	234	600	150
	5	Katarivog	233	600	150
	6	BR28	230	600	150
	7	BR29	233	600	150
	8	Paijam	234	600	150
	9	Bashful	252	600	150
	10	Lal Aush	231	600	150
	11	Jirashail	231	600	150
	12	Gutisharna	256	600	150
	13	Red Cargo	240	600	150
	14	Najirshail	231	600	150
	15	Katari Polao	242	600	150
	16	Lal Biroi	231	600	150
	17	Chinigura Polao	237	600	150
	18	Amon	250	600	150
	19	Shorna5	238	600	150
	20	Lal Binni	232	600	150
Dataset-02 [35]	21	Arborio	15000	600	150
	22	Turkish Basmati	15000	600	150
	23	Ipsala	15000	600	150
	24	Jasmine	15000	600	150
	25	Karacadag	15000	600	150
Dataset-03 [63]	26	BD30	500	600	150
	27	BD33	500	600	150
	28	BD39	500	600	150
	29	BD49	500	600	150
	30	BD51	500	600	150
	31	BD52	500	600	150
	32	BD56	500	600	150
	33	BD57	500	600	150
	34	BD70	500	600	150
	35	BD72	500	600	150
	36	BD75	500	600	150
	37	BD76	500	600	150
	38	BD79	500	600	150
	39	BD85	500	600	150
	40	BD87	500	600	150
	41	BD91	500	600	150
	42	BD93	500	600	150
	43	BD95	500	600	150
	44	Binadhan7	500	600	150
	45	Binadhan8	500	600	150
	46	Binadhan10	500	600	150
	47	Binadhan11	500	600	150
	48	Binadhan12	500	600	150
	49	Binadhan14	500	600	150
	50	Binadhan16	500	600	150
	51	Binadhan17	500	600	150
	52	Binadhan19	500	600	150
	53	Binadhan21	500	600	150
	54	Binadhan23	500	600	150
	55	Binadhan24	500	600	150
	56	Binadhan25	500	600	150
	57	Binadhan26	500	600	150
	58	BR22	500	600	150
	59	BR23	500	600	150
	60	BRRI67	500	600	150
	61	BRRI74	500	600	150
	62	BRRI102	500	600	150
			98230	**37200**	**9300**

study, four CNN backbones has been selected for transfer learning: VGG16, MobileNetV2, InceptionV3, and Xception. These architectures are chosen for their complementary strengths in feature extraction, computational efficiency, and classification performance, which are further demonstrated in the ablation study.

Figs. 2–5 illustrate the architectural designs of the selected CNN backbones: VGG16, MobileNetV2, InceptionV3, and Xception respectively. In Table 3, a comparative summary discussion of the applied 4 different fine-tuned transfer-learning models with relevance of usage to our rice varieties classification is given.

3.2.2. Meta-Ensemble learning approach

The proposed architecture, illustrated in Fig. 6, leverages a stacked-ensemble learning framework for rice variety classification. Initially, the rice dataset undergoes feature extraction through four optimized DL backbone models, generating four distinct feature sets $\{F_1, F_2, F_3, F_4\}$. Each feature set represents a high-dimensional embedding of the rice images, where convolutional or transformer-based encoders capture shape, texture, and color attributes. Formally, if $x_i \in R^d$ is an input rice image, the feature extraction process can be expressed as:

$$F_j = f_j(x_i; \theta_j), \quad j \in \{1, 2, 3, 4\}$$

Where, $f_j(\cdot)$ is the jth DL model parameterized by θ_j, and F_jDenotes the resulting feature representation. These feature sets are then aggregated and passed into a meta-learning stage. In this stage, 13 meta-classifiers are trained on the stacked outputs. The meta-classifiers employed in this study include Logistic Regression, K-Nearest Neighbours (KNN), Support Vector Classifier (SVC), Decision Tree Classifier, Random Forest Classifier, Adaptive Boosting (AdaBoost) Classifier, Extreme Gradient Boosting (XGB) Classifier, Gaussian Naïve Bayes (Gaussian NB), Passive Aggressive Classifier, Ridge Classifier, Stochastic Gradient Descent (SGD) Classifier, Extremely Randomized Trees (Extra Trees) Classifier, and the Multi-Layer Perceptron (MLP) Classifier. These meta-learners are intentionally selected to represent a wide spectrum of learning paradigms ranging from linear and margin-based methods to probabilistic models, tree-based ensembles, and neural approaches. This diversity ensures that each classifier contributes a distinct inductive bias, enabling the ensemble to capture different structural properties within the deep feature space. By leveraging these complementary strengths, the stacked model reduces variance, improves generalization, and achieves higher robustness compared to any individual CNN or transformer model.

If we denote the predictions from the base learners as $\widehat{y_1}, \widehat{y_2}, \ldots, \widehat{y_m}$, the stacked ensemble combines them into a new feature vector:

$$Z = [\widehat{y_1}, \widehat{y_2}, \ldots, \widehat{y_m}]$$

Where, Z is the meta-level input representation. The meta-classifier $g(\cdot)$, parameterized by ϕ, learns to map these combined predictions to the final class label $\widehat{y}$:

$$\widehat{y} = g(Z; \phi)$$

This hierarchical structure allows the model to capture complementary strengths of different DL backbones, while the meta-classifiers learn an optimal decision boundary from the stacked features. The final prediction corresponds to the best-performing meta-classifier selected during evaluation, yielding the most reliable classification output.

3.2.3. Vision transformers

The ViT adapts the transformer architecture, originally designed for Natural Language Processing (NLP), to visual data. Instead of directly processing the entire image, ViT divides an image into fixed-size patches and treats them as sequential tokens, analogous to words in a sentence. Fig. 7 shows the general architecture of our ViT model.

In our raw ViT, the rice grain image is first divided into non-overlapping fixed-size patches. Given an image $x \in R^{H \times W \times C}$It is split into $N = \frac{HW}{P^2}$ patches, each of size $P \times P$ with C channels. Each flattened patch $x_p^i \in R^{P^2 C}$ is linearly projected into a D-dimensional embedding space through:

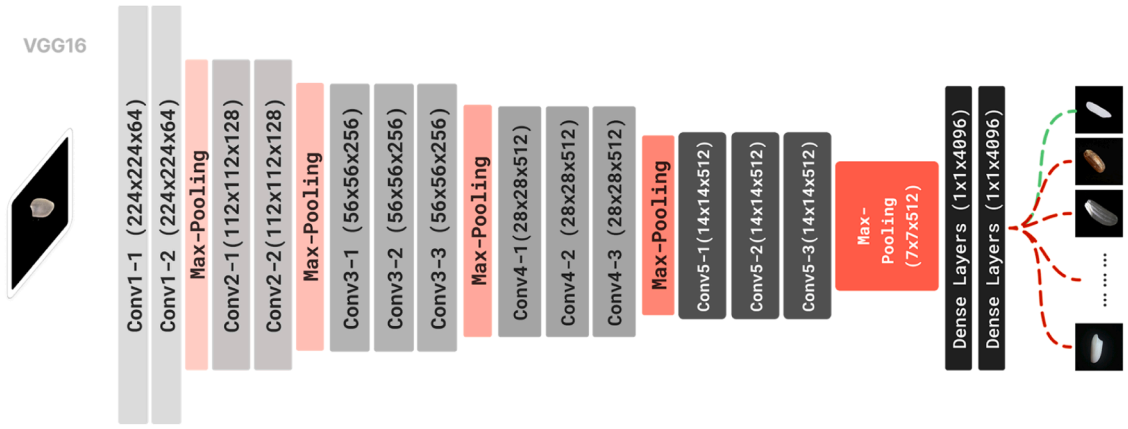


Fig. 2. Fine-tuned VGG16 model's architecture.

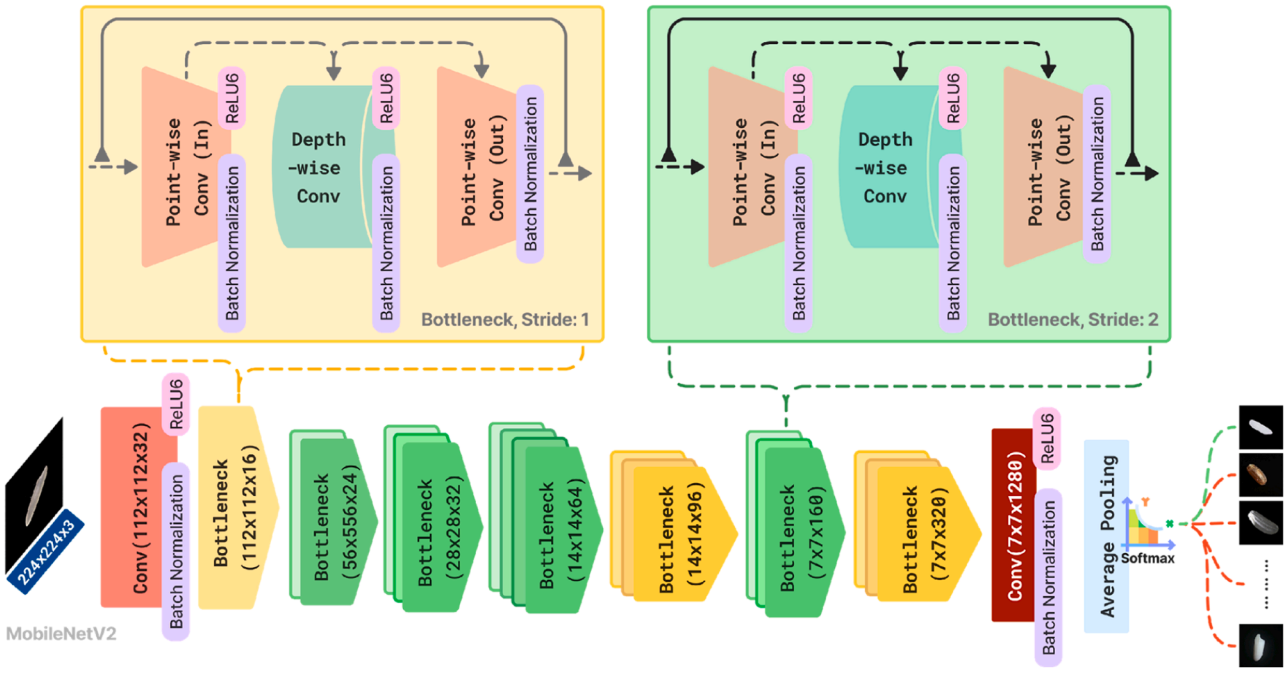


Fig. 3. Fine-tuned MobileNetV2 model's architecture.

$$z_0^i = x_p^i W_E + b_E, \quad W_E \in R^{(P^2C)\times D}$$

To maintain spatial ordering, a learnable positional embedding $E_{pos} \in R^{(N+1)\times D}$ is added to the patch embeddings, along with a special classification token x_{class}:

$$z_0 = \left[x_{class}; z_0^1; z_0^2; \ldots; z_0^N\right] + E_{pos}$$

This sequence passes through L stacked Transformer encoder layers, where each layer contains Multi-Head Self-Attention (MHSA) and a Feed-Forward Network (FFN).

The self-attention mechanism computes similarity among patches using queries (Q), keys (K), and values (V):

$$Attention(Q, K, V) = Softmax!\left(\frac{QK^T}{\sqrt{d_k}}\right)V,$$

Where, $Q = XW^Q, K = XW^K, V = XW^V$. For multi-head attention with h heads:

$$MHA(X) = [head_1, \ldots, head_h]W^O.$$

The FFN applies two linear transformations with GELU activation:

$$FFN(x) = GELU(xW_1 + b_1)W_2 + b_2.$$

Residual connections and layer normalization are applied:

$$z_l' = z_{l-1} + MHA(LN(z_{l-1})), \quad z_l = z_l' + FFN\left(LN(z_l')\right).$$

Finally, the classification token z_L^0 is passed through a softmax classifier:

$$y = Softmax\left(z_L^0 W_c + b_c\right).$$

This raw ViT is highly dependent on dataset size and tuning. Without large pre-training, performance may be unstable, but it effectively demonstrates the role of attention in visual recognition tasks.

On the other hand, our fine-tuned Google's ViT-B/16 follows the same core pipeline as the raw ViT but introduces standardized architectural choices and large-scale pre-training. In this model, the image is divided into 16×16 patches ($P = 16$), resulting in $N = \frac{HW}{16^2}$ tokens. Each patch is linearly embedded into a 768-dimensional vector, with a learnable classification token and sinusoidal/learnable positional

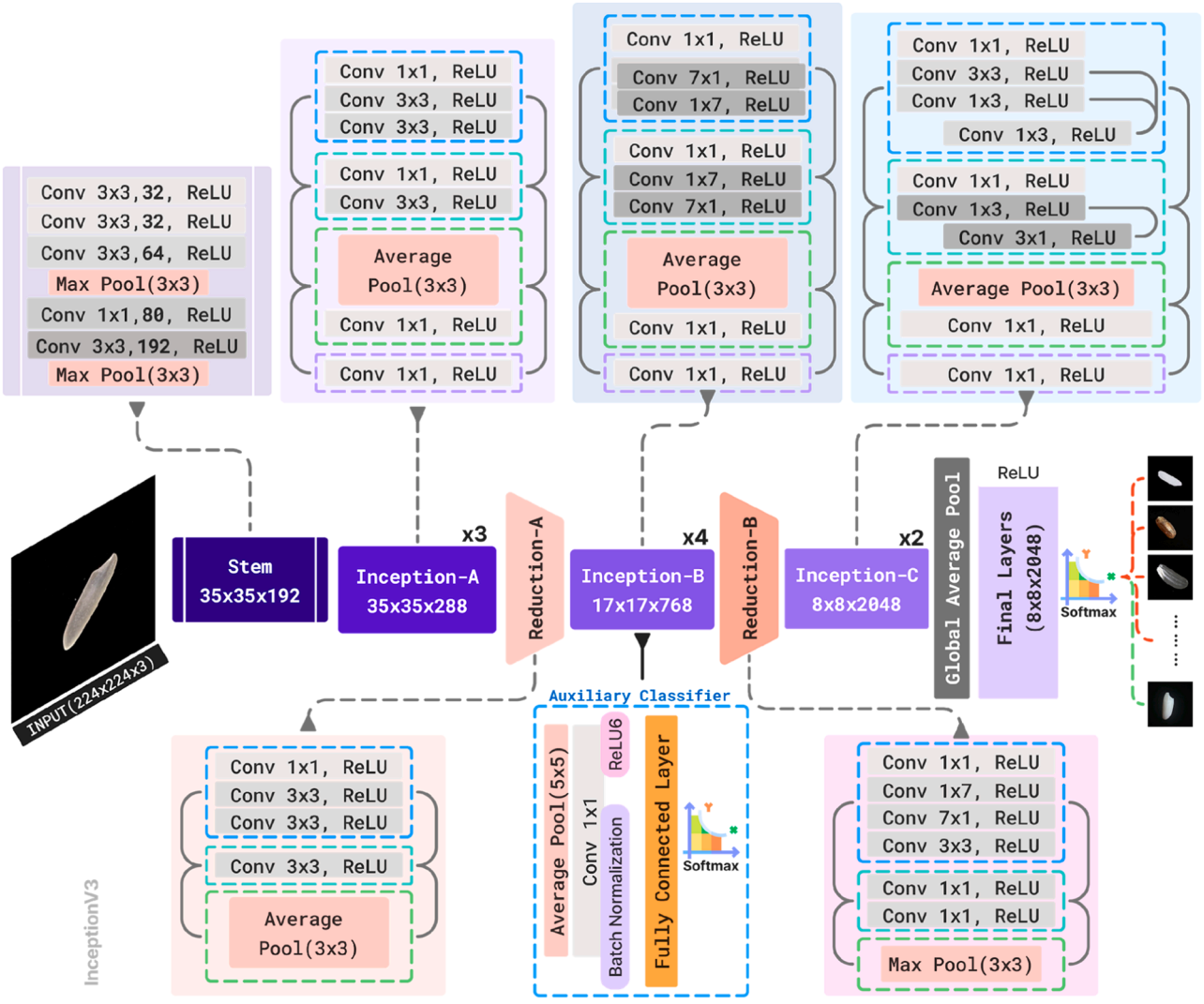


Fig. 4. Fine-tuned InceptionV3 model architecture.

encodings added. The encoder consists of 12 Transformer layers, each with 12 attention heads and hidden size $D = 768$. The MHSA mechanism is identical:

$$Attention(Q, K, V) = Softmax!\left(\frac{QK^T}{\sqrt{d_k}}\right)V,$$

But with a much larger capacity due to multiple attention heads and higher dimensionality. The FFN expands dimensions to 3072 before projecting back to 768:

$$FFN(x) = GELU(xW_1 + b_1)W_2 + b_2, \quad W_1 \in R^{768\times3072}.$$

The final classification token embedding is passed into a linear classifier trained with cross-entropy loss. Since Google ViT-B/16 is pre-trained on ImageNet-21k (14 M images, 21k classes) and fine-tuned on ImageNet-1k, it transfers effectively to new domains like rice variety classification, providing superior generalization compared to raw ViT.

3.2.4. Model deployment

The deployment process of our models has been designed to ensure accessibility across both web-based and mobile platforms, making the model usable for researchers, farmers, and industry professionals. The deployment pipeline begins with the trained rice classification model, which is developed using DL architectures and optimized for performance. For lightweight mobile deployment readiness, the selected best-performing VGG16 model is first converted into TensorFlow Lite (TFLite) format, ensuring reduced model size and computational efficiency, which is crucial for running on mobile and embedded devices with limited hardware resources. On other hand, for web app deployment, three different best performing models (VGG16, stacked-ensemble, and Google's ViT) is deployed separately. Fig. 8 illustrates the whole Web and Mobile App deployment process.

- **RiceVision WebApp** [65]: On the web application, it handles incoming requests; processes uploaded rice grain images and returns predictions in real time. Users interact with the WebApp through a simple interface, where they upload a single rice grain image (minimum 5x zoom to ensure feature clarity). The backend system then runs inference using the deployed model and outputs the predicted rice variety name along with additional information. This web interface provides a platform-independent solution, accessible from any browser without requiring local installation.
- **RiceVision Mobile App** [66]: For the mobile application, the TFLite-converted model is integrated directly into an Android application through Android Studio. The app provides an intuitive interface where users can capture or upload rice grain images and instantly obtain predictions on-device. By embedding the model

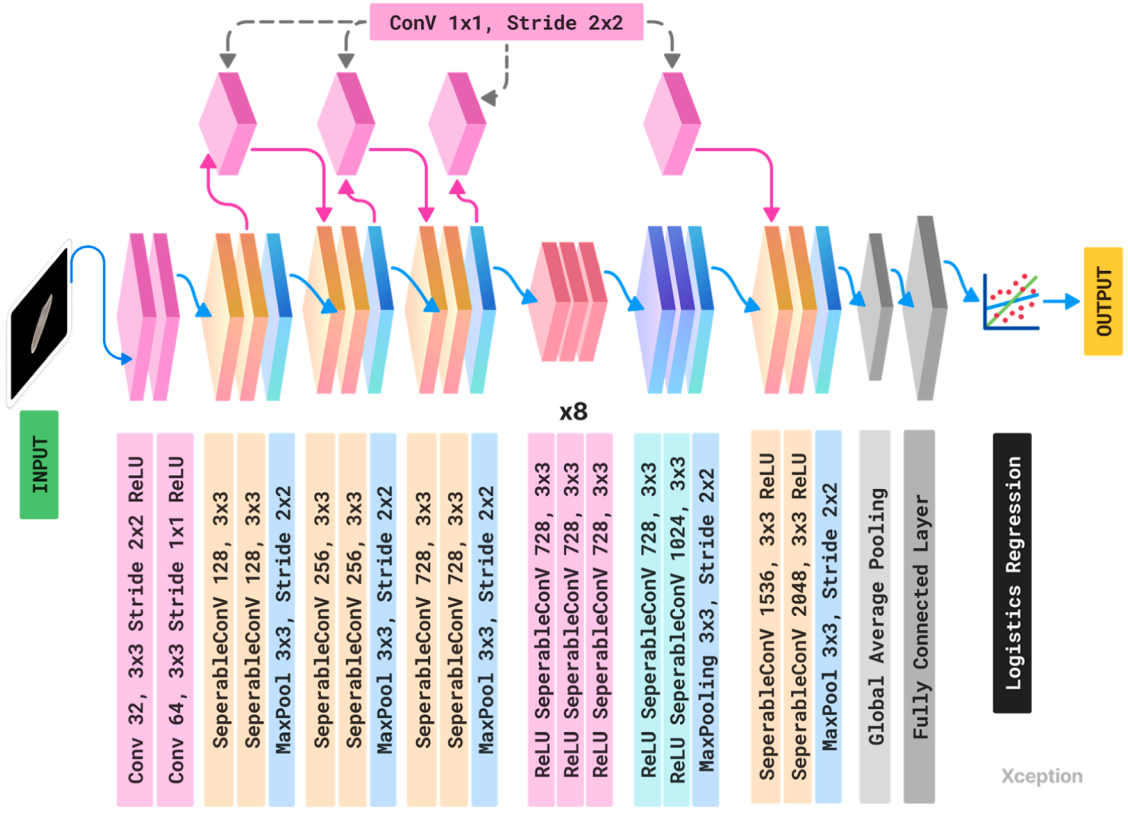


Fig. 5. Fine-tuned Xception model architecture.

Table 3
Architectural justification of selected CNN models for rice variety classification.

Model	Architectural Nature	Strengths	Relevance to Rice Classification
VGG16	Deep sequential 3 × 3 convolutional layers with max-pooling	Captures fine-grained local spatial patterns; stable hierarchical feature extraction	Effectively learns grain edges, shape, and surface details important for differentiating visually similar varieties
MobileNetV2	Depthwise separable convolutions with inverted residuals	Lightweight, fast, and memory-efficient while maintaining strong feature representation	Suitable for deployment in web/mobile systems; captures subtle texture and colour variations in rice grains
InceptionV3	Parallel convolutions with multiple kernel sizes (multi-scale feature extraction)	Learns both fine details and global structural patterns simultaneously	Distinguishes varieties based on grain length, width, and texture variations across classes
Xception	Depthwise separable convolutions replacing Inception modules	Efficiently separates spatial and channel-wise feature learning	Enhances discrimination of subtle texture and reflectance differences across 62 rice varieties

locally, predictions can be made without requiring internet connectivity, which is particularly valuable in remote agricultural regions.

3.3. Experimental setup and hyper-parameter tuning

Developing an effective classification model typically involves three key phases: (i) development, where hyperparameters are selected; (ii) training and validation, where model parameters are optimized; and (iii) evaluation on unseen data. The success of this process depends both on the diversity of the dataset and on the careful choice of hyperparameters such as learning rate (α), batch size, dropout rate, number of epochs, and optimizer. While the training set is used to guide parameter updates, the validation set provides a safeguard against overfitting and informs hyperparameter refinement.

In this study, experiments have been implemented in TensorFlow/Keras and executed on an NVIDIA Tesla V100 GPU with 32 GB of memory. The dataset has been partitioned into 80 % training and 20 % testing, with class balance preserved. CNN backbones (VGG16, MobileNetV2, InceptionV3, and Xception) have been initialized with ImageNet-pretrained weights and adapted by replacing the top layers with a global average pooling layer, fully connected dense layers, dropout regularization, and a *softmax* classifier for 62 classes. Training followed a two-stage procedure:

i. **Feature extraction** – backbone layers have frozen, and only the classification head has been trained using the Adam optimizer.
ii. **Fine-tuning** – the top 30–40 % of backbone layers have unfrozen, enabling domain-specific feature adaptation with a reduced learning rate.

For transformer models, we examined both a raw ViT trained from scratch and the pretrained Google ViT-B/16, fine-tuned on our dataset. The optimized hyperparameter settings for CNN and ViT models are summarized in Tables 4 and 5, respectively. These configurations provided the foundation for the experimental evaluations reported in the following section.

The selection of hyperparameters is based on an initial sensitivity analysis shown in Table 6, which demonstrated that moderate batch sizes (32), a cosine-decayed learning rate, and a dropout of 0.3 provided the most stable convergence across CNN backbones. For Vision

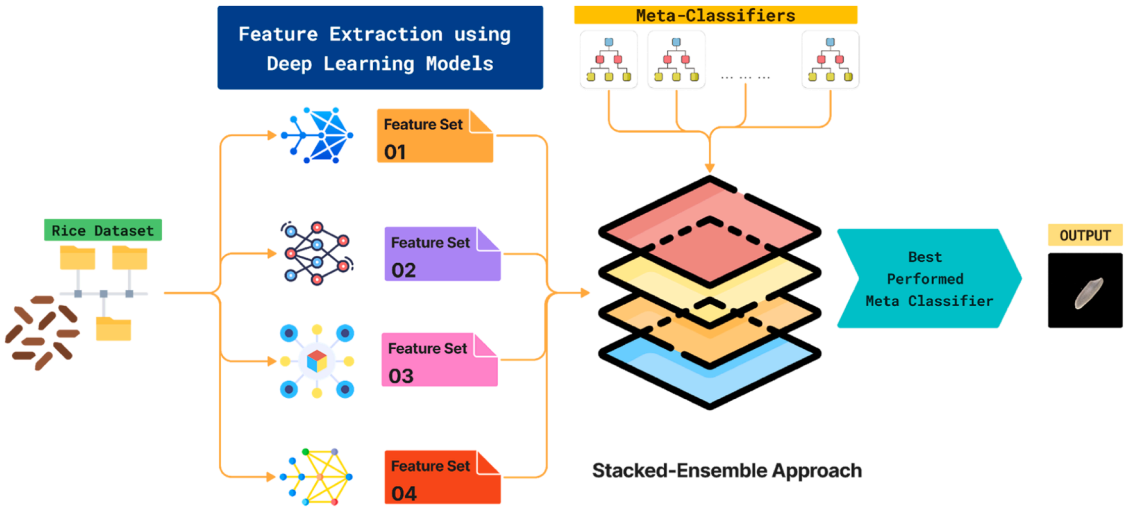


Fig. 6. Meta-Ensemble method architecture.

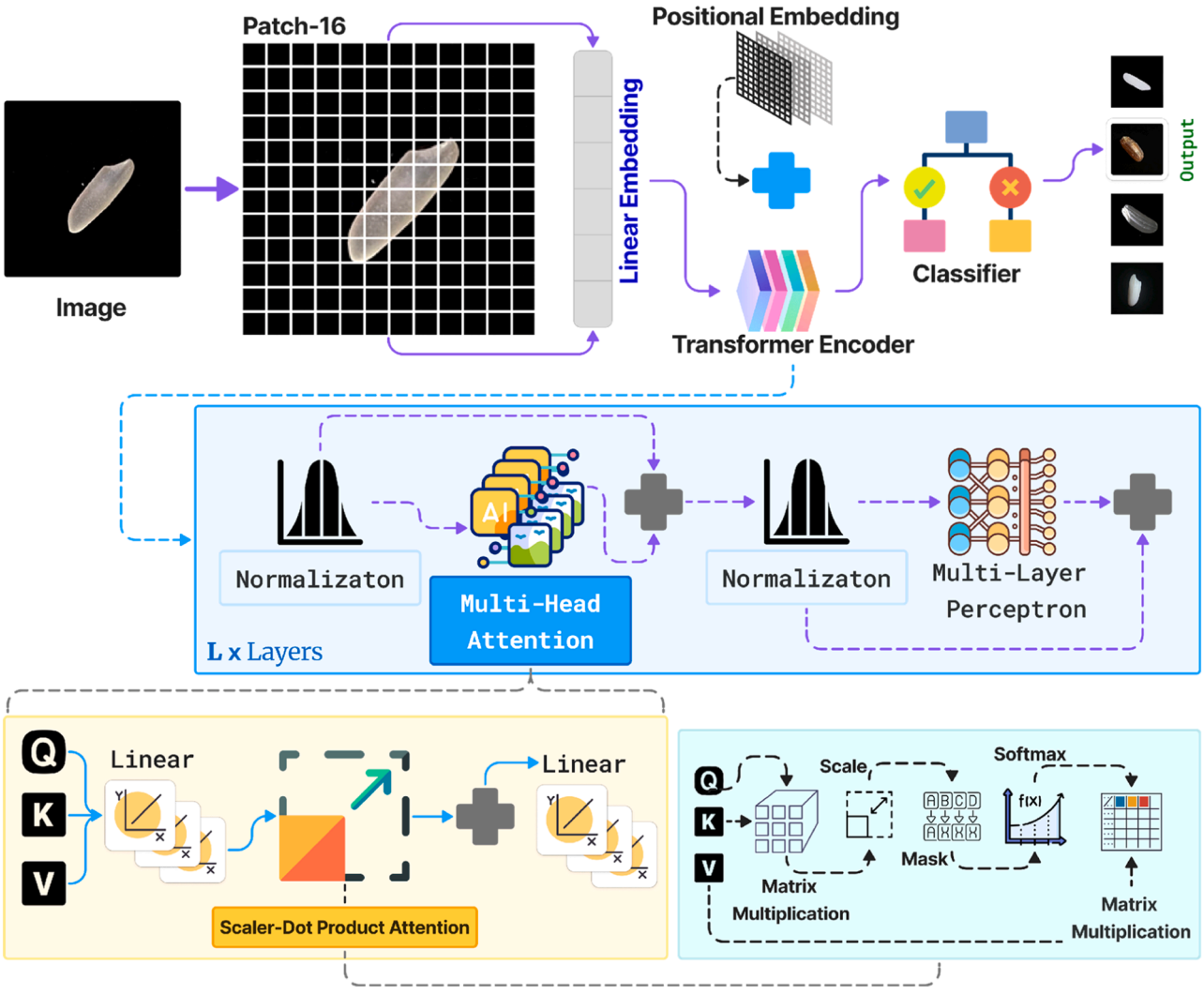


Fig. 7. ViT model's basic architecture.

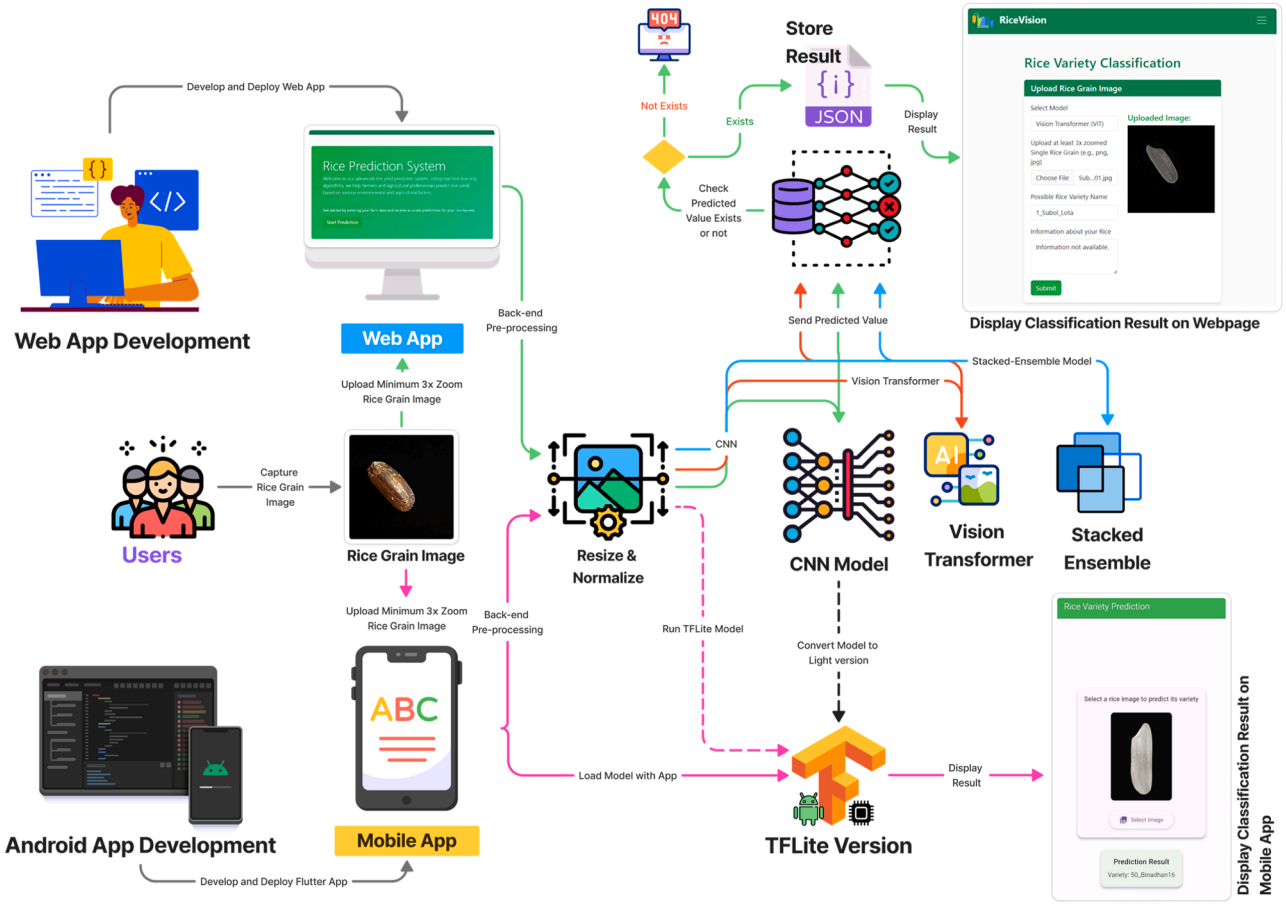


Fig. 8. Deployment framework of *RiceVision* on web and mobile platforms.

Table 4
Optimized Hyperparameter settings for transfer-learning models.

Parameter	Setting
Image Size	224 × 224 (RGB)
Batch Size (per replica)	16 (scaled to global batch = 16 × replicas)
Backbones Tested	MobileNetV2, VGG16, InceptionV3, Xception
Pretraining	ImageNet weights (transfer learning)
Head Layers	GAP → Dropout(0.30) → Dense(256, ReLU, L2=1e-4) → BatchNorm → Dense(62, Softmax)
Dropout Rate	0.3
Regularization	L2 = 1e-4 (applied to Dense layer)
Loss Function	Categorical Cross-Entropy with Label Smoothing = 0.05
Optimizer (Stage 1)	Adam, Learning Rate = 3×10^{-4}
Optimizer (Stage 2)	Adam + Cosine Decay (initial LR = 1×10^{-4}, α = 0.1)
Training Stages	Stage 1: Train head only (8 epochs) Stage 2: Fine-tune top 40 % backbone (25 epochs)
Unfreezing Policy	Top 40 % of backbone layers unfrozen in Stage 2
Callbacks	EarlyStopping (patience=8, monitor=val_acc) ReduceLROnPlateau (patience=3, factor=0.2, min_lr=1e-6) ModelCheckpoint (weights only, best val_acc)
Precision Policy	Mixed Precision (bfloat16, logits in float32 for stability)

Transformers, a patch size of 16 and the AdamW optimizer has been chosen due to their superior stability and reduced risk of attention collapse for mid-scale datasets such as ours.

Table 5
Optimized Hyper-parameter Settings for ViT-Raw and google's ViT-B/16 models.

Parameter	ViT (Raw Implementation)	Google's ViT-b/16 Implementation
Image Size	224 × 224	224 × 224
Patch Size	16	16
Batch Size	32	32
Number of Classes	62	62
Epochs	Variable (15–30)	30
Learning Rate (LR)	0.001 with CosineDecay	0.0001 (AdamW)
Weight Decay	0.01 (Dynamic)	0.1
Optimizer	Adam (Cosine Decay LR scheduler)	AdamW + CosineAnnealingLR
Dropout Rate	0.3 (applied in transformer encoder and MLP)	0.3
Projection/Embed Dim	256	1024
MLP Hidden Dim	512	512
Attention Heads	8	16
Transformer Layers	4	12
Regularization	L2 (1e-4 to 1e-6)	AdamW weight decay
Early Stopping	Yes (patience=5)	Not Applied
Scheduler	Cosine Decay	CosineAnnealingLR

Table 6
Sensitivity Analysis of Hyperparameters.

Hyperparameter	Tested Range	Optimal	Reason
Batch Size	16,32,64,128	32	Balance gradient stability and GPU memory
LR (CNN)	3e-4 to 1e-5	3e-4 → 1e-4 decay	Fast initial learning + fine-tuning stability
LR (ViT)	1e-3 to 1e-5	1e-4 (AdamW)	Prevent attention layer collapse
Dropout	0.2, 0.4, 0.6	0.3	Reduces overfitting in FC layers
Patch Size	8 and 16	16	Best trade-off between locality and context

3.4. Performance evaluation metrics

To rigorously evaluate the performance of rice variety classification models, several standard classification metrics are employed. These metrics provide complementary insights into the predictive capabilities of the models.

1. Accuracy: Accuracy measures the proportion of correctly classified samples among the total samples.

$$Accuracy = TP + TNTP + TN + FP + FNAccuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

where TP = True Positives, TN = True Negatives, FP = False Positives, and FN = False Negatives.

2. Precision: Precision reflects the model's ability to correctly identify positive samples out of all predicted positives.

$$Precision = TPTP + FPPrecision = \frac{TP}{TP + FP}$$

A high precision indicates a low false positive rate.

3. Recall: Recall (or sensitivity) measures the proportion of actual positive samples that are correctly identified.

$$Recall = TPTP + FNRecall = \frac{TP}{TP + FN}$$

A high recall indicates that most actual positives are correctly captured by the model.

4. F1-Score: The F1-score is the harmonic mean of precision and recall, balancing both metrics.

$$F1 - Score = 2 \times Precision \times RecallPrecision + RecallF1 - Score = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

This is particularly useful in imbalanced datasets where accuracy alone may be misleading.

5. Confusion Matrix: The confusion matrix provides a detailed breakdown of predictions across all classes. For a multi-class classification problem with C classes, it is represented as a $C \times C$ matrix. where c_{ij} denotes the number of samples belonging to class i that are predicted as class j. The diagonal entries represent correct classifications, while off-diagonal entries indicate misclassifications.

6. Receiver Operating Characteristic (ROC) Curve: The ROC curve illustrates the trade-off between the True Positive Rate (TPR) and False Positive Rate (FPR) at various decision thresholds.

$$TPR\ (Recall) = \frac{TP}{TP + FN}, \quad FPR = \frac{FP}{FP + TN}$$

By plotting TPR against FPR, the ROC curve shows the discriminative power of the classifier. The Area Under the Curve (AUC) quantifies overall performance, where a value closer to 1 indicates superior classification ability.

4. Results and discussion

Results are organized into five subsections: (i) transfer learning backbones, (ii) stacked ensemble strategies, (iii) transformer-based models (raw ViT and pretrained ViT-B/16), (iv) comparative analysis across methods, and (v) an ablation study examining the influence of architecture, hyperparameter tuning, and ensemble design on model robustness and generalization. Model performance is reported using accuracy and F1-score, while confusion matrices and ROC curves provide class-level insights into strengths, weaknesses, and misclassification patterns.

4.1. Transfer learning results

A comprehensive overview of the performance of four CNN backbones on our dataset is provided in Table 7. All models achieved training accuracy of 1.00 % with low training losses, confirming effective feature learning. However, differences emerged in validation performance. VGG16 attained the best accuracy (91.13 %) with balanced precision, recall, and F1-score (≈0.91), while MobileNetV2 achieved comparable accuracy (91.01 %) but lower consistency across metrics (0.84). By contrast, InceptionV3 (88.26 %) and Xception (88.12 %) produced slightly lower accuracies and higher validation losses.

The Figs. 9–11 present the training and validation accuracy/loss curves, the confusion matrix, and the ROC curves of the fine-tuned VGG16 model, which achieved the best performance among the CNN backbones. The confusion matrix demonstrates a highly accurate classifier, with a strong diagonal pattern indicating correct predictions across almost all 62 rice varieties. Misclassifications are extremely sparse and concentrated within a few fine-grained varietal groups rather than dispersed randomly. The zoomed-in region highlights this pattern clearly: varieties such as BD33, BD39, BD51, BD52, and BD75 show small off-diagonal values, reflecting occasional confusion. These varieties share nearly identical grain morphology—similar length–width ratios, translucency, and dorsal chalk patterns—making them visually challenging even for human inspectors. The multi-class ROC curves show that the classifier maintains excellent discriminative power across almost all 62 rice varieties, with AUC values largely between 0.98 and 1.00. The curves cluster near the top-left corner, indicating high true-positive rates at very low false-positive rates. Slightly lower AUC values correspond to the same fine-grained varieties noted in the confusion matrix, confirming that any performance drop arises from inherent visual similarity rather than model weakness. Overall, the ROC results demonstrate strong sensitivity, robust class separation, and reliable performance across diverse rice types.

4.2. Misclassification analysis

According to VGG16's confusion matrix as shown in Fig. 10 most

Table 7
Accuracy and loss of applied CNN models.

Sl No.	Model	Train Accuracy	Train Loss	Validation Accuracy	Validation Loss	Precision	Recall	F1-Score
1	VGG16	1.0000	0.4906	0.9113	0.7607	0.9200	0.9100	0.9100
2	MobileNetV2	0.9990	0.5133	0.9101	0.7732	0.8400	0.8400	0.8400
3	InceptionV3	1.0000	0.4687	0.8826	0.8541	0.8800	0.8800	0.8800
4	Xception	1.0000	0.4600	0.8812	0.8407	0.8800	0.8800	0.8800

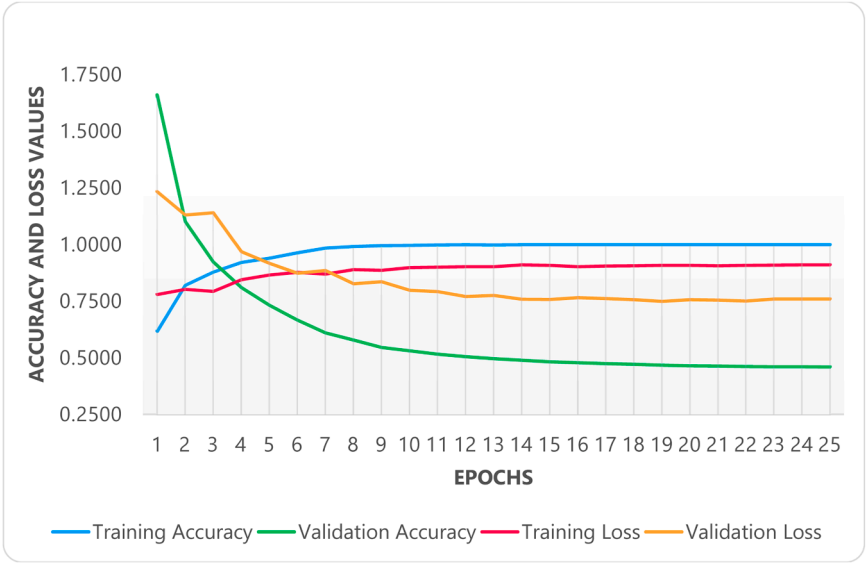


Fig. 9. Training and testing accuracy and loss curves of the best-performing CNN Model (VGG16).

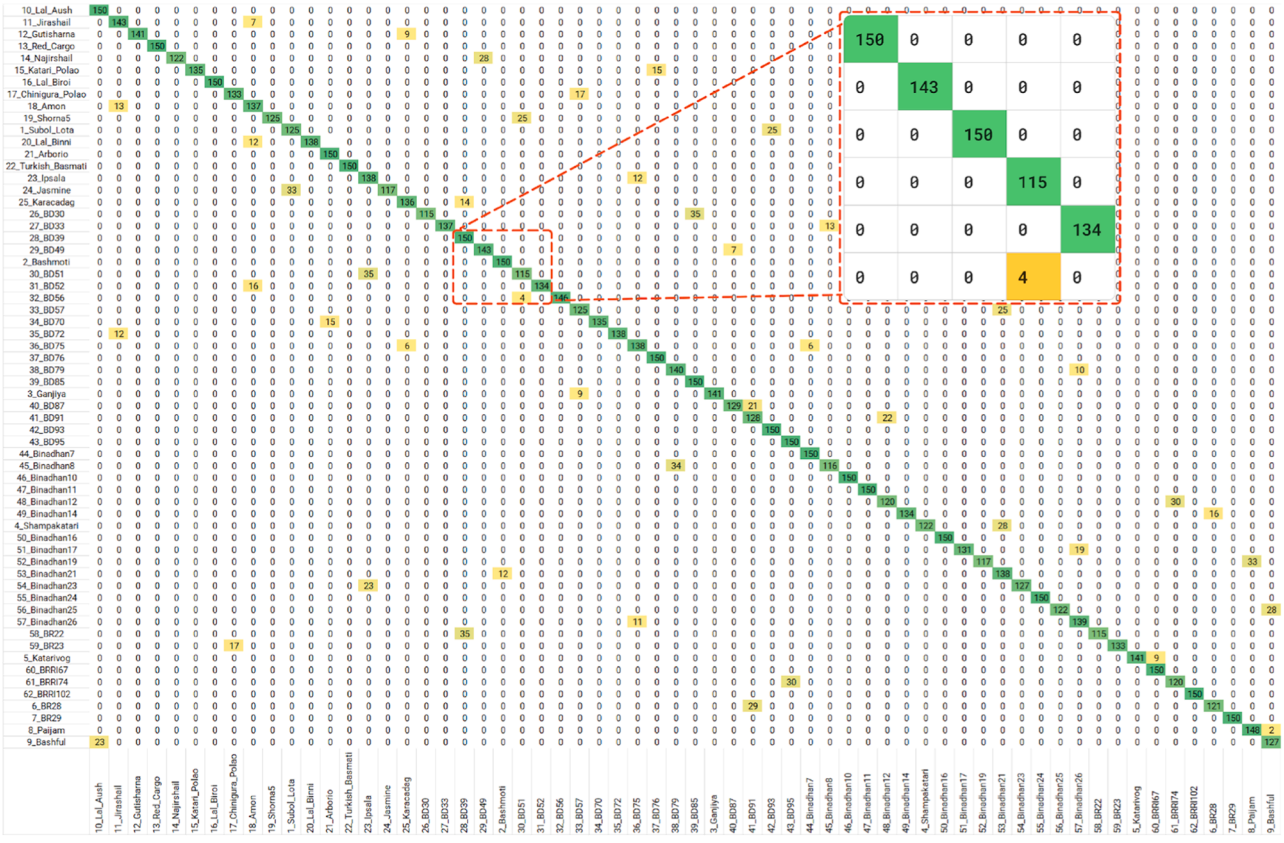


Fig. 10. Confusion matrix of the best-performing CNN model (VGG16).

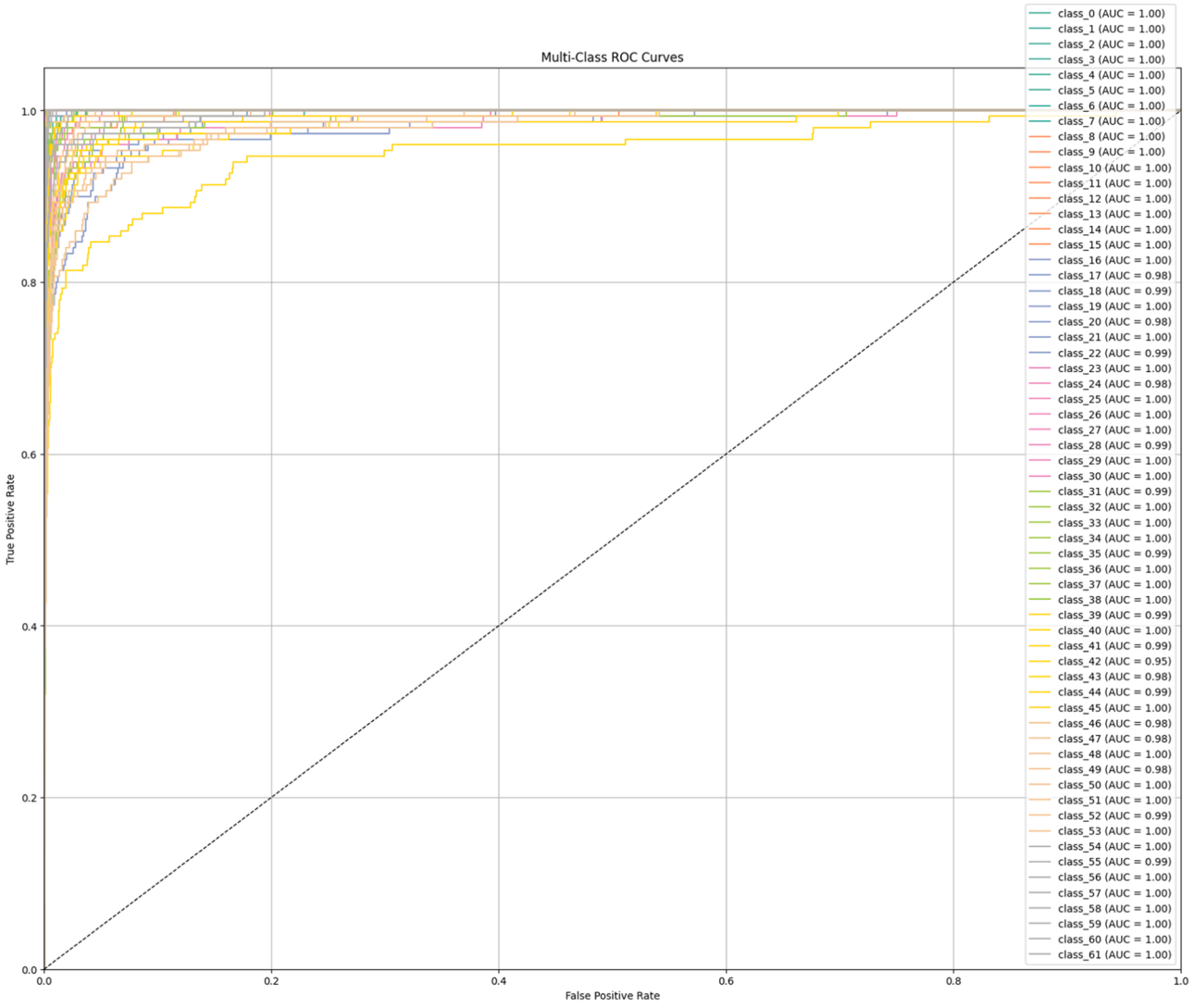


Fig. 11. ROC curve of the best-performing CNN model (VGG16).

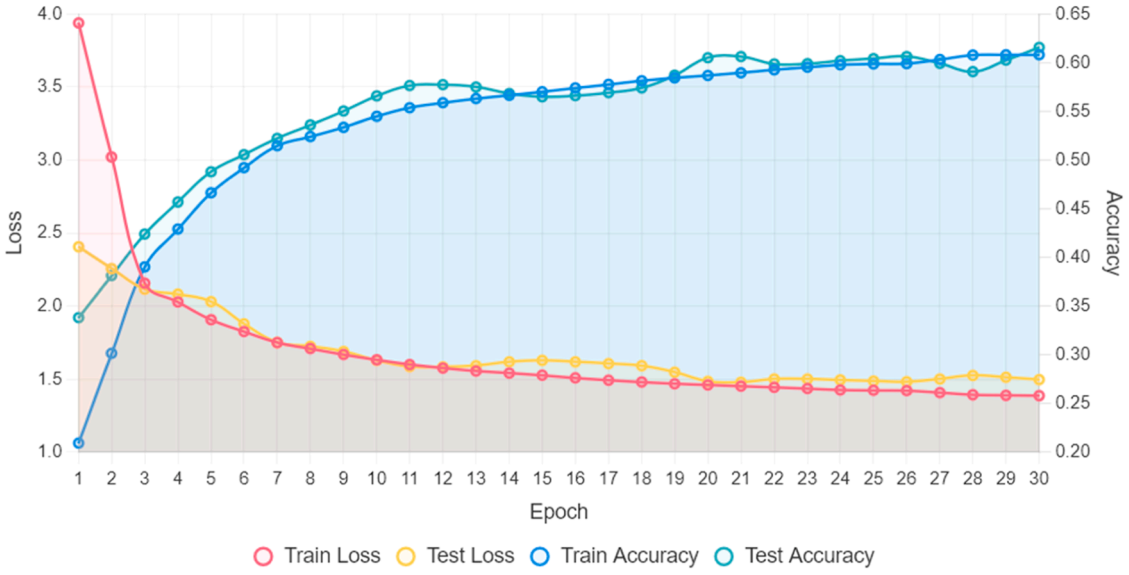


Fig. 12. Training and testing accuracy and loss curves of the raw ViT model.

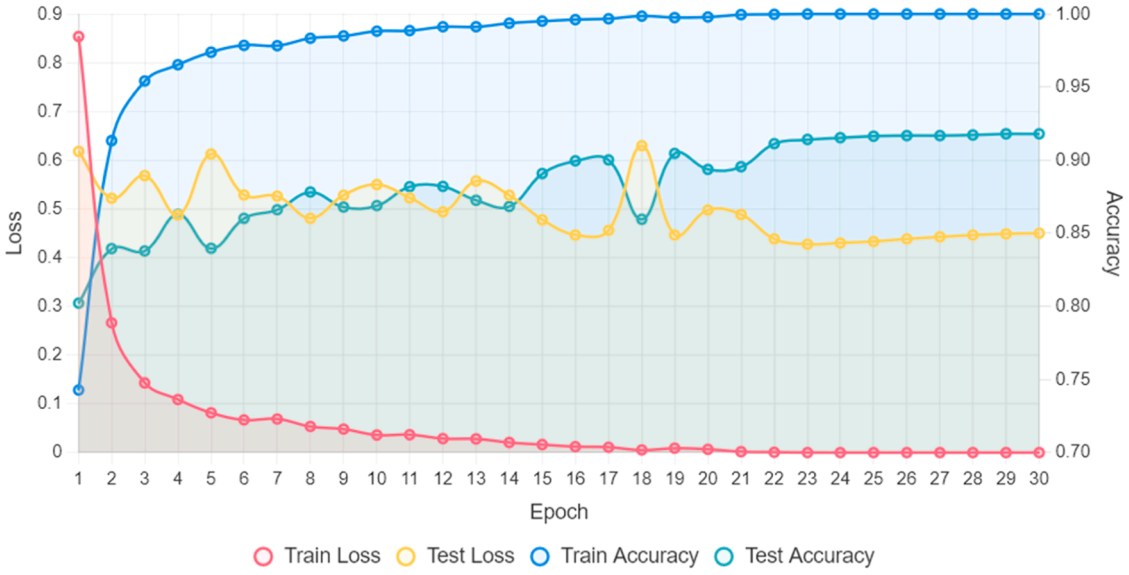


Fig. 13. Training and testing accuracy and loss curves of Google's ViT-B/16 model.

misclassifications occur within closely related varietal groups, where kernels share highly similar morphological and textural traits. These errors are predominantly confined to BD-series and Binadhan-series varieties, which often exhibit nearly identical grain length, width, translucency, and dorsal-line structure such as BD33↔BD35, BD39↔BD38, BD51↔BD52, and Binadhan16↔Binadhan17. These groups also belong to the same breeding lineage, where milling patterns and internal chalk distribution are inherently similar, further reducing discriminative features. Importantly, the confusion matrix shows no widespread cross-class errors; instead, mistakes are localized within morphologically homogeneous families. This indicates that the model has learned strong class boundaries overall, with remaining errors arising from biologically challenging look-alike grains rather than model weakness. Strengthening inter-class separability within these clusters through targeted augmentation, hard-negative mining, or contrastive learning would likely reduce these specific misclassification patterns.

Table 9
Accuracy and loss of raw ViT and Pretrained ViT models.

Model (Backbone)	Train Accuracy	Train Loss	Validation Accuracy	Validation Loss
Raw ViT	0.6122	1.3740	0.6156	1.4966
Google's ViT-B/16	1.0000	0.0000	0.9180	0.4501

4.3. Stacked ensemble results

The Table 8 presents the accuracies of stacked ensemble models evaluated with 13 meta-classifiers across three CNN feature combinations. The results demonstrate that ensemble learning provides consistently high performance, with most classifiers achieving accuracies above 0.96. Tree-based methods, including Decision Tree (DT), Random Forest (RF), Extra Trees (ET), and XGBoost (XGB), reached the highest accuracy of 1.000, reflecting their capacity to effectively partition complex feature spaces and exploit the complementary representations extracted from multiple CNN backbones. Linear models such as Logistic

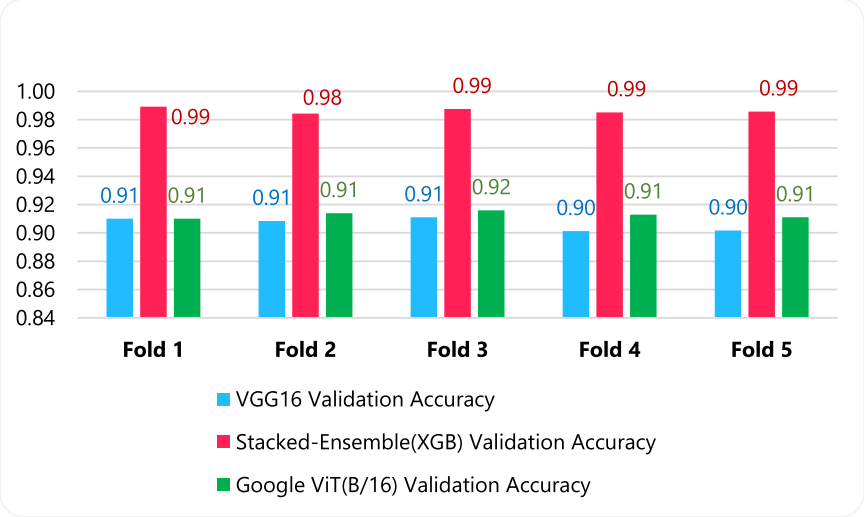


Fig. 14. Robustness analysis showing consistent 5-fold validation accuracy for VGG16, stacked-ensemble (XGBoost), and Google ViT-B/16, demonstrating stable generalization across all folds.

Regression (LR), Support Vector Classifier (SVC), and Ridge Classifier (RC) also produced strong results (≈0.97–0.99), showing that even simpler learners achieve near-optimal performance when provided with rich deep feature sets. In contrast, AdaBoost exhibited instability, performing poorly in the two-model stack (0.5205) but improving substantially (>0.99) with larger ensembles, highlighting its dependence on feature diversity. Probabilistic and margin-based classifiers (Gaussian Naïve Bayes and Passive Aggressive) showed slightly lower but still competitive performance (≈0.91–0.99), while the Multi-Layer Perceptron (MLP) remained consistent across all combinations (≈0.96–0.98). Overall, these findings confirm that stacked ensembles are highly effective for the dataset, with tree-based meta-classifiers and diverse CNN feature inputs delivering the most reliable and robust classification outcomes.

4.4. *ViT's results*

The performance of the two ViT models is summarized in Table 8. The raw ViT achieved a training accuracy of 61.22 % and a validation accuracy of 61.56 %, with relatively high training and validation losses. In contrast, the pretrained Google ViT-B/16 model attained 100 % training accuracy and 91.80 % validation accuracy. In the case of the raw ViT as shown in Fig. 12, the training and test accuracies improve gradually, reaching around 62 % by the end of 30 epochs. However, the loss curves show relatively high fluctuations, with test loss stabilizing but not achieving low values, indicating limited generalization capacity. This reflects the challenge of training ViT models from scratch on rice grain images, as they require very large datasets and longer training times to reach competitive accuracy.

In contrast, Google's ViT-B/16 as shown in Fig. 13 demonstrates a much stronger performance. The training accuracy quickly converges to nearly 100 %, while test accuracy stabilizes above 90 %, reflecting excellent feature extraction and generalization capability. The training and test losses are significantly lower and more stable compared to the raw ViT, showing the effectiveness of pre-training on large-scale datasets like ImageNet and subsequent fine-tuning. In our experiments, the raw ViT model showed weaker performance because ViTs require large-scale datasets to learn patch-level relationships from scratch. With mid-sized data such as our rice variety dataset, the model struggles to form stable attention patterns, leading to underfitting and high loss. In contrast, the pretrained ViT-B/16 benefits from extensive ImageNet pretraining, allowing it to transfer well-structured visual representations to the rice classification task, resulting in significantly higher accuracy and robustness.

Table 9 highlights the train and test accuracy & loss values for our applied optimized raw ViT and Google's ViT-B/16 model.

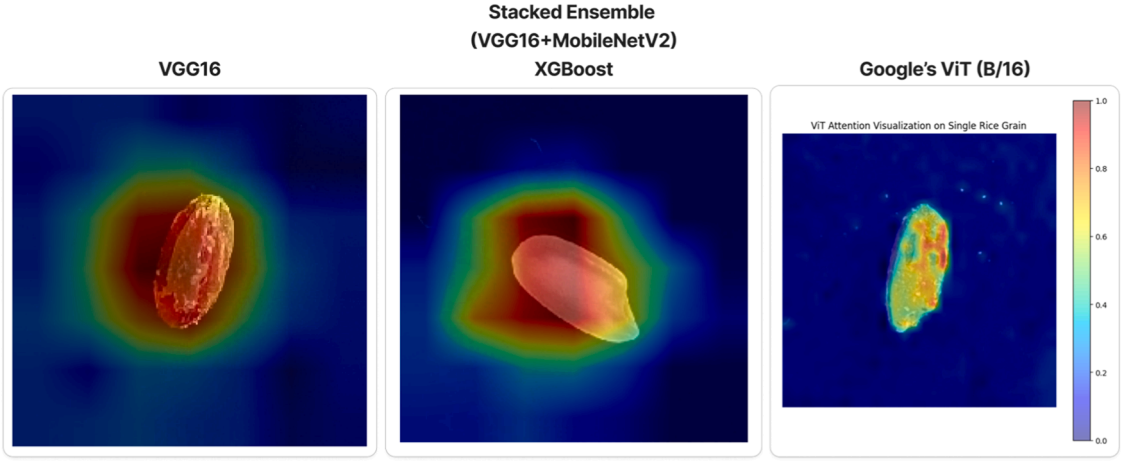


Fig. 15. Explainability visualizations for rice kernel classification using three model types.

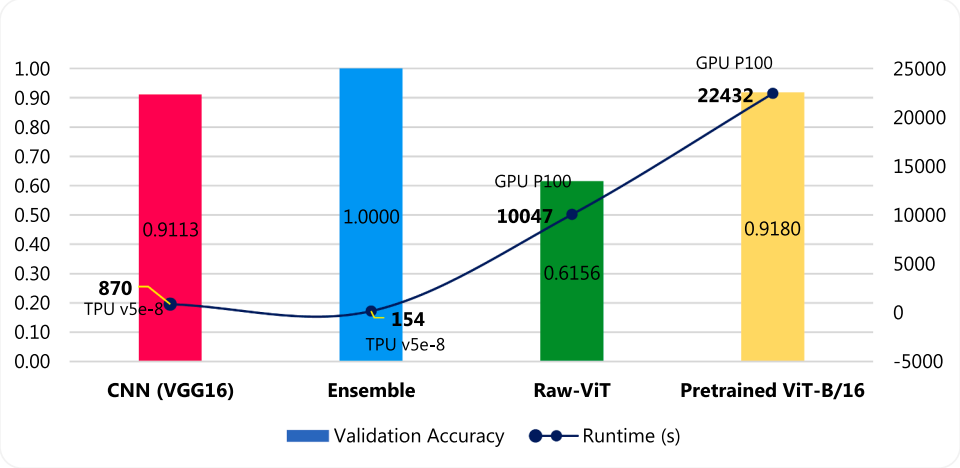


Fig. 16. Comparative accuracy of different deep-learning approaches along with runtime for rice variety classification.

4.5. Robustness analysis

Fig. 14 demonstrates the stability and reliability of the three evaluated models across 5-fold cross-validation. VGG16 shows consistently strong performance, with accuracies ranging from 0.90 to 0.91 across all folds, indicating excellent generalization and minimal variance. The stacked ensemble with XGBoost maintains high accuracy between 0.98 and 0.99, confirming that its near-perfect performance is not due to data leakage or overfitting but is reproducible across different data splits. In contrast, the pretrained Google ViT-B/16 model remains stable around 0.91–0.92 in every fold, reflecting consistent transformer behavior on this fine-grained dataset. Together, these results verify that all three models exhibit high robustness, with performance variations remaining minimal across the folds, further strengthening the reliability of the proposed framework.

Table 10
Comparative analysis of rice variety classification studies highlighting dataset size, number of classes, applied methodologies, performance results, and practical applicability.

Study	Dataset Size	Number of Classes	Methodology	Result (%)	Real-Word Application
Saxena et al. [40]	75000	5	Random Forest	99.85	×
Iqbal et al. [45]	75000	5	CNN	99.20	×
Tasci et al. [44]	75000	5	Quantized Neural Network	99.87	×
Setiawan et al. [47]	45000	3	Ensemble Learning	96.74	×
Kang et al. [46]	550	5	Principal Component Analysis, BP Neural Network, Random Forest	95.30	×
Islam et al. [49]	27,000	20	Ensemble Approach	100	✔
Islam et al. [50]	27000 6750	20 5	ConvNext Models	99.60 100	×
Proposed	46500	62	VGG16 Ensemble Approach ViT-Google-B/16	91.13 99.99 91.80	✔

4.6. Explainability analysis

Fig. 15 visualization methods reveal how different architectures focus on discriminative patterns within a single rice kernel. VGG16 Grad-CAM highlights broad shape- and texture-based cues, concentrating mainly on the central body of the grain, which suggests strong reliance on macroscopic morphology. The ensemble heatmap (VGG16 + MobileNetV2) produces a more refined and stable activation region, capturing both coarse structure and subtle surface irregularities—indicating that stacked learning effectively aggregates complementary spatial features. In contrast, the ViT-B/16 attention map provides the most detailed and fine-grained focus, distributing attention across multiple micro-patches rather than a single region. This reflects the transformer's ability to capture long-range relationships and variation in local patterns along the grain surface. Together, these explainability results show that while CNNs emphasize global shape, the ensemble balances global and local cues, and ViT models extract higher-resolution structural signatures—offering a more nuanced understanding of why the model succeeds in distinguishing visually similar rice varieties. On other hand, GradCAM and Guided GradCAM along with ViT attention map visualization across all 62 types of rice kernel sample are shown in Appendix A (Fig. F1 and Fig. F2) consecutively.

4.7. Comparative analysis

Fig. 16 compares the performance of all methods/models on the dataset. The stacked ensemble achieved the highest accuracy (≈100 %), demonstrating the benefit of combining CNN feature sets with strong meta-classifiers such as XGBoost. Google's ViT-B/16 also performed strongly (91.80 %), surpassing the single CNN backbone VGG16 (91.13 %). In contrast, the raw ViT achieved only 61.56 %, underscoring the importance of pretraining and fine-tuning for transformer architectures in fine-grained agricultural classification.

The computational analysis compares validation accuracy with model runtime to highlight the trade-off between performance and computational cost. Among all models, VGG16 achieved strong accuracy (0.9113) with relatively low runtime (870 s) on TPU, demonstrating high efficiency. The stacked ensemble produced the highest accuracy (1.0000) while also being the most computationally economical (154 s), making it the optimal choice for practical deployment. In contrast, the raw ViT required substantially higher computation (10,047 s on GPU) and still yielded lower accuracy (0.6156), confirming the limitations of training transformers from scratch on mid-scale datasets. The pretrained ViT-B/16 significantly improved accuracy (0.9180) but incurred the highest computational cost (22,432 s), reflecting the heavy complexity of transformer architectures. Overall, the results indicate that while

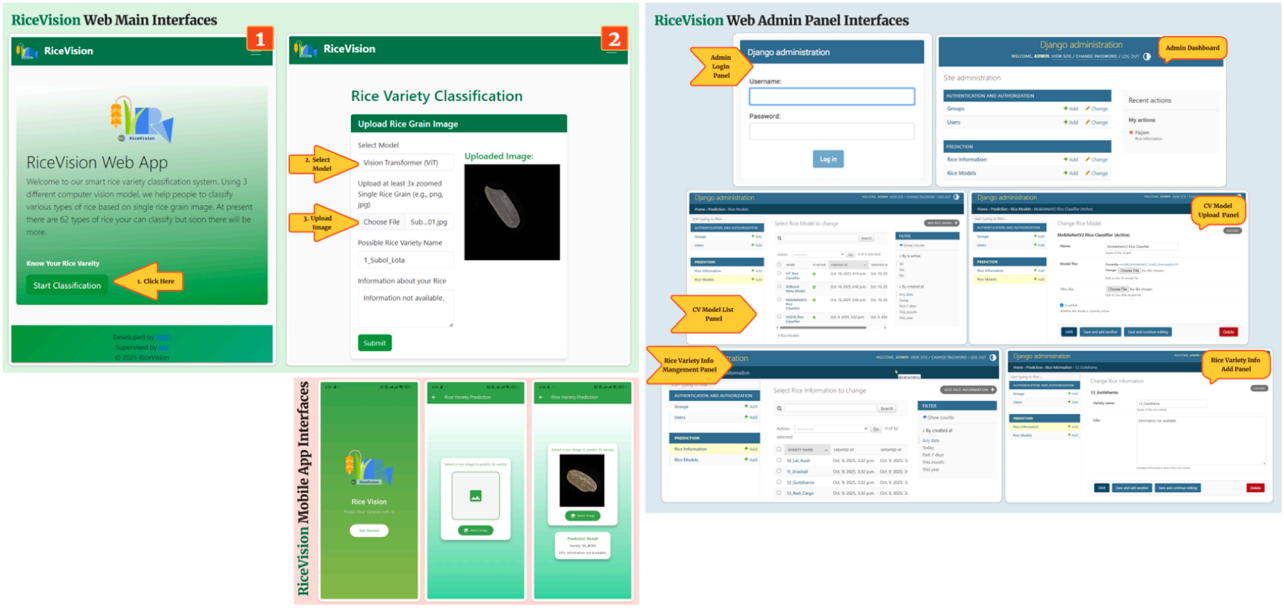


Fig. 17. RiceVision web and mobile application interfaces.

pretrained transformers offer strong performance, CNN-based models and ensemble fusion achieve the best balance between accuracy and computational efficiency, making them more suitable for real-time agricultural applications.

Table 10 summarizes prior rice variety classification studies across datasets, class sizes, and methodologies. Unlike earlier studies [40, 44–47,49,50] that used small and limited datasets covering only 3–20 rice varieties, our work introduces the largest and most diverse benchmark to date with 62 classes and 46,500 images, collected and standardized across multiple verified sources. This scale enables true fine-grained recognition that previous models could not achieve. Methodologically, we move beyond single-model CNN approaches by integrating complementary deep features through a stacked ensemble architecture, which significantly enhances generalization and minimizes model-specific bias. We further extend prior work by incorporating pretrained transformer models, offering a comprehensive cross-architecture comparison that has not been previously explored in rice classification. Most importantly, our framework is translated into real-world practice through deployment on both web and mobile platforms, demonstrating practical readiness for seed authentication, industry applications, and research use an aspect largely absent in earlier studies.

4.8. Web and mobile application

RiceVision is an open-source, cross-platform system designed for real-time rice variety identification through deep learning. The web application [65], developed in Django, enables users to upload single-grain images directly through a browser. The interface includes two modules: the User Panel, which supports image upload, automatic preprocessing (resize, RGB normalization), and model selection among fine-tuned VGG16, ensemble, or Vision Transformer (ViT-B/16); and the Admin Panel, which allows authorized users to manage datasets, update models, monitor logs, and maintain rice-variety metadata through a structured dashboard. Predictions are generated within seconds and displayed with the predicted variety name.

The mobile application [66], developed in Flutter and integrated with a quantized TensorFlow Lite model (~30 MB), performs fully offline inference, making it suitable for field use. Users can capture or select images, which the app pre-processes locally before running on-device predictions. Results appear instantly with minimal latency and full data privacy. Fig. 17 shows the RiceVision Web and Mobile App interfaces.

5. Ablation study and insights

To evaluate the effect of architectural choices and optimization strategies, extensive ablation experiments have been conducted by varying model backbones, hyperparameters, and ensemble configurations. The detailed results are summarized in Table 11, which reports validation accuracy and loss under different configurations, and in Table 12, which highlights key insights derived from these experiments. For CNN backbones, fine-tuning with different batch sizes, dropout rates, and learning-rate schedules revealed consistent trends. VGG16 achieved the most stable convergence across settings, while MobileNetV2 balanced accuracy and efficiency. In contrast, deeper models such as ResNet50 and NasNetMobile showed reduced convergence and higher losses, confirming that excessive depth can hinder performance on mid-sized and comprehensive datasets. For transformer-based models, the raw ViT trained from scratch underperformed due to patch-size sensitivity and limited data, as reflected in Table 10. By comparison, the pretrained Google ViT-B/16 yielded substantially higher accuracy, validating the importance of transfer learning for adapting attention-based models to agricultural classification. As detailed in Table 11, tree-based learners (RF, ET, XGBoost) consistently reached the highest accuracy, highlighting their ability to exploit complementary CNN representations and reduce variance. In summary, the ablation experiments confirm three critical insights: (i) moderate-depth CNNs with tuned hyperparameters outperform deeper architectures, (ii) pretrained transformers are competitive when supported by large-scale transfer learning, and (iii) ensemble fusion remains the most reliable strategy for maximizing performance in fine-grained rice classification.

5.1. Limitations

Our combined dataset contains several visually indistinguishable varieties, especially within BD and Binadhan families, making fine-grained separation inherently difficult. Images has been captured under controlled single-kernel conditions, which may limit real-world generalization. Some classes show low intraclass variation, and subtle

Table 11
Summary of ablation experiments highlighting the impact of hyperparameter tuning.

Model	Patch Size	Weight Decay	Epochs	Pre-Trained	Image Size	Batch Size	Learning Rate	Optimizer	Dropout Rate	L2 Regularization	Loss Function	Validation Accuracy	Validation Loss
Google ViT B/16	16	0.1	30	Yes	224	8	0.0001 – 0.0003	AdamW	0.3	Not Applied	Categorical Cross-Entropy	0.9100	0.6664
	16	0.1	30	Yes	224	32	0.0001 – 0.0003	AdamW	0.3	Not Applied	Categorical Cross-Entropy	0.9059	0.8675
Raw ViT	14	0.01	30	No	224	8	0.001	AdamW	0.3	0.01	Categorical Cross-Entropy	0.5500	1.6955
	8	0.01	30	No	224	8	0.001	AdamW	0.3	0.01	Categorical Cross-Entropy	0.6126	1.0041
MovileNetV2	Not Applicable	Not Applicable	25	Yes	224	16	0.0003–0.0001	Adam	0.3	0.0001	Categorical Cross-Entropy	0.9088	0.7865
	Not Applicable	Not Applicable	25	Yes	224	32	0.0003–0.0001	Adam	0.3	0.0001	Categorical Cross-Entropy	0.8956	0.9741
	Not Applicable	Not Applicable	25	Yes	224	64	0.0003–0.0001	Adam	0.3	0.0001	Categorical Cross-Entropy	0.8941	0.9915
ResNet50	Not Applicable	Not Applicable	25	Yes	224	16	0.0003–0.0001	Adam	0.3	0.0001	Categorical Cross-Entropy	0.7869	1.1967
	Not Applicable	Not Applicable	25	Yes	224	32	0.0003–0.0001	Adam	0.3	0.0001	Categorical Cross-Entropy	0.7930	1.1157
	Not Applicable	Not Applicable	25	Yes	224	64	0.0003–0.0001	Adam	0.3	0.0001	Categorical Cross-Entropy	0.7962	1.6527
MobileNet	Not Applicable	Not Applicable	30	Yes	224	32	0.0001-0.00001	Adam AdamW	0.5	Not Applied	Categorical Cross-Entropy	0.9102	1.0612
	Not Applicable	Not Applicable	30	Yes	224	64	0.0001-0.00001	Adam AdamW	0.5	Not Applied	Categorical Cross-Entropy	0.9034	1.5228
	Not Applicable	Not Applicable	30	Yes	224	16	0.0001-0.00001	Adam AdamW	0.5	Not Applied	Categorical Cross-Entropy	0.9042	1.4897
ResNet50V2	Not Applicable	Not Applicable	30	Yes	224	32	0.0001–0.00001	Adam AdamW	0.5	Not Applied	Categorical Cross-Entropy	0.8663	1.1692
	Not Applicable	Not Applicable	30	Yes	224	64	0.0001–0.00001	Adam AdamW	0.5	Not Applied	Categorical Cross-Entropy	0.8758	1.1045
	Not Applicable	Not Applicable	30	Yes	224	16	0.0001–0.00001	Adam AdamW	0.5	Not Applied	Categorical Cross-Entropy	0.8717	1.0470
NasNetMobile	Not Applicable	Not Applicable	25	Yes	224	32	0.0001–0.00001	Adam AdamW	0.5	Not Applied	Categorical Cross-Entropy	0.7987	1.4042
	Not Applicable	Not Applicable	25	Yes	224	64	0.0001–0.00001	Adam AdamW	0.5	Not Applied	Categorical Cross-Entropy	0.7973	1.4407
	Not Applicable	Not Applicable	25	Yes	224	16	0.0001–0.00001	Adam AdamW	0.5	Not Applied	Categorical Cross-Entropy	0.8004	1.2570

Table 12
Insights of various experiments highlighting the effect of architecture, hyperparameter tuning, and ensemble strategies on rice classification performance.

Factor / Setting	Variation Tested	Observed Effect
Architecture	CNN backbones	CNNs showed strong performance, leveraging hierarchical feature extraction.
	Raw ViT	Struggled due to data-hungry nature, coarse patching, and lack of pretraining.
	Pretrained ViT-B/16	Significant improvement and transfer learning boosted performance close to the CNN level.
Hyperparameters	Learning rate (1e-3, 1e-4, schedulers)	Fine-tuning with a smaller LR improved convergence stability.
	Dropout (0.3–0.5)	Prevented overfitting, and optimal dropout is around 0.3–0.4.
	Batch size (16–64)	Moderate batch sizes balanced stability and generalization.
	Optimizers (Adam, SGD, RMSprop)	Adam gave faster convergence, and SGD with momentum yielded stable results in CNNs.
Training Strategy	Stage-wise fine-tuning vs. full network training	Stage-wise fine-tuning yielded higher validation accuracy and reduced overfitting.
	Early stopping	Prevented over-training, stabilized validation loss.
Ensemble Methods	Stacked ensemble of 2, 3, and 4 CNN models	Ensembles consistently outperformed single models.
	Meta-learners	Some meta-learners, such as XGB, RF, reached ~100 % accuracy, boosting ensemble robustness.
Deployment Consideration	VGG16 Ensemble Model Google's ViT (B/16)	User can choose which model to classify rice variety. These have the best accuracy that made it suitable for app deployment.

features can be diluted by ViT patching. Finally, the model relies solely on visual cues without integrating physical or biochemical attributes that could further improve discrimination.

6. Conclusions

Accurate rice variety classification is a critical requirement for breeding programs, seed authentication, and modern supply chain management, where subtle differences among cultivars must be reliably distinguished. This work represents the first unified attempt to build a multi-source, expert-verified, and standardized rice variety dataset at this scale. The integrated benchmarking of CNNs, ensembles, and transformers, combined with rigorous cross-validation and field-tested deployment, offers a comprehensive foundation for future research in grain informatics and agricultural AI. Using this dataset, we systematically evaluated CNN backbones, stacked ensemble strategies, and transformer-based models. The stacked ensemble achieved the highest accuracy by integrating complementary CNN features and reducing model-specific variance, while VGG16 and pretrained ViT-B/16 emerged as strong single-model baselines. To demonstrate translational value, the best performed models from CNN (VGG16), ensemble, and ViT all are deployed in our web-based platform and only VGG16 is deployed in lightweight mobile application (RiceVision), enabling real-time classification for research, seed certification, and industrial use cases. This study illustrates how deep learning models can be adapted for practical deployment in agriculture. Future work will prioritize dataset augmentation with multi-grain and field-captured samples, along with exploration of federated learning for collaborative yet privacy-preserving model training.

Data availability statement

The data supporting the findings of this study are available in a public repository: https://www.kaggle.com/datasets/abrahametry/rice-62-varieties-dataset

Conflict of interest disclosure

The authors have no relevant financial or non-financial interests to disclose. The authors have no competing interests to declare that are relevant to the content of this article.

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Ethical statement

This study does not involve any experiments on humans or animals, and no ethical approval was required. Therefore, an ethical statement is not applicable.

CRediT authorship contribution statement

Md. Masudul Islam: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Md. Shafiqul Islam:** Writing – review & editing, Visualization, Supervision. **Md. Golam Moazzam:** Validation, Supervision, Project administration. **Mohammad Shorif Uddin:** Writing – review & editing, Validation, Supervision, Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Supplementary materials

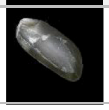
Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.atech.2025.101710.

Appendix A

Table A1's Legend of Feature Categories

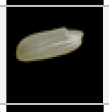
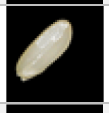
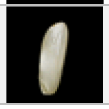
- **Shape:** Long / Medium / Short / Slender / Broad / Curved / Straight
- **Tip Morphology:** Sharp / Blunt / Tapered / Hooked / Rounded
- **Surface Texture:** Smooth / Rough / Striated / Patchy / Chalky / Bran-Colored
- **Color / Transparency:** Translucent / Opaque / Semi-Opaque / White / Cream / Brownish / Bran-coated
- **Distinctive Marks:** Chalk patches, grooves, dents, wrinkles, embryo spot shape, etc.

Table A1
Visual Feature Comparison of All Rice Varieties.

Variety	Image	Shape	Tip Morphology	Surface Texture	Color / Transparency	Distinctive Visual Notes
AMON		Medium–short	Slight hook	Patchy + chalk spots	Semi-opaque	Brownish spots on surface
ARBORIO		Short, round	Blunt	Very smooth	White, highly opaque	Classic plump Arborio type
BASHFUL		Short	Rounded	Smooth	Light translucent	Slight belly curvature
BD30		Long	Tapered	Mild striations	Translucent	Smooth dorsal line
BD33		Long	Tapered	Smooth	Semi-translucent	Visible center chalk band
BD39		Long	Rounded tip	Mildly chalky	Semi-translucent	Wispy chalk traces
BD49		Long	Sharp	Striated	Semi-opaque	Greyish tint, dorsal ridge
BD51		Long	Sharp	Rough patches	Semi-translucent	Broad dorsal chalk line
BD52		Long	Sharp	Rough	Semi-opaque	Dense white patch at center
BD56		Long	Slight taper	Smooth	Grey-translucent	Mild stripe marks
BD57		Long	Sharp	Smooth	Translucent	Faint vertical groove
BD70		Long	Sharp	Smooth	Slightly opaque	White embryo spot
BD72		Long	Narrow tip	Patchy	Greyish	Multiple chalk spots
BD75		Long	Sharp	Striated	Semi-opaque	Scattered chalk pits
BD76		Long	Tapered	Rough	Opaque	Heavy chalk regions
BD79		Long	Sharp	Rough	Semi-translucent	Pale greenish hue

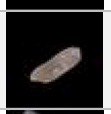
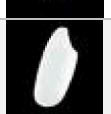
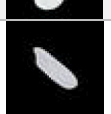
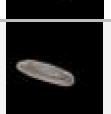
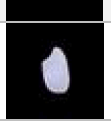
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Table A1 (*continued*)

BD85		Long	Ridge-sharp	Smooth	Semi-translucent	Wrinkled appearance
BD87		Medium	Blunt	Smooth	Semi-translucent	Slight dorsal crease
BD91		Medium–long	Tapered	Smooth	Light opaque	Soft chalk patches
BD93		Long	Rounded	Smooth	Semi-opaque	Greyish tone
BD95		Long	Tapered	Smooth	Light opaque	Mild belly chalk
BINADHAN07		Long	Sharp	Smooth	Opaque	Subtle chalk bands
BINADHAN08		Long	Sharp	Rough	Semi-translucent	Oval chalk mark at center
BINADHAN10		Long	Tapered	Smooth	Opaque-white	Strong dorsal chalk stripe
BINADHAN11		Long	Sharp	Smooth	Translucent	Fine vertical grain pattern
BINADHAN12		Long	Slight hook	Pitted	Semi-opaque	Large chalk blotch area
BINADHAN14		Long	Sharp	Smooth	Semi-translucent	Thin, slender body
BINADHAN16		Long	Hooked	Smooth	Semi-opaque	Slight embryo dent
BINADHAN17		Long	Sharp	Striated	Opaque	Deep dorsal crease
BINADHAN19		Long	Very slender	Smooth	Translucent	Needle-like shape
BINADHAN21		Long	Sharp	Smooth	Semi-translucent	Uniform matte texture
BINADHAN23		Medium	Blunt	Slightly chalky	Semi-opaque	Roundish belly
BINADHAN24		Medium–long	Tapered	Smooth	Semi-translucent	Light tip chalk

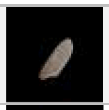
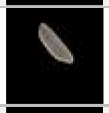
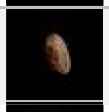
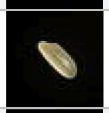
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Table A1 (*continued*)

BINADHAN25		Long	Sharp	Rough	Semi-translucent	Dense chalk clusters
BINADHAN26		Long	Rounded	Smooth	Semi-opaque	Soft surface marks
BR22		Long	Sharp	Smooth	Semi-translucent	Very clear dorsal ridge
BR23		Long	Sharp	Smooth	Slightly opaque	Tip has chalk bloom
BR28		Medium	Blunt	Smooth	Semi-translucent	Curved belly
BR29		Medium–long	Blunt-taper	Smooth	Semi-translucent	Mild chalk belly
BRRI67		Long	Sharp	Smooth	Semi-transparent	Dense center chalk
BRRI74		Long	Wide-belly	Smooth	Greyish	Round embryo pit
BRRI102		Long	Sharp	Smooth	Semi-opaque	Long dorsal chalk streak
CHINIGURA POLAO		Short	Round	Smooth	White opaque	Small classic aromatic grain
DESHI BASHMOTI		Medium-long	Sharp	Smooth	Light opaque	Slight aromatic-type curvature
GANJIYA		Very short	Blunt	Brown husk	Bran-colored	Only variety with full bran visible
GUTISHORNA		Medium	Blunt	Patchy	Semi-opaque	Large chalk + husk remnants
IPSALA		Long	Sharp	Smooth	Translucent	Turkish type, very slender
JASMINE		Medium-short	Blunt	Smooth	White opaque	Classic jasmine appearance
JIRASHAIL		Medium	Tapered	Smooth	Semi-opaque	Slight chalk lines
KARACADAG		Long	Blunt	Patchy	Opaque	White thick chalk layer

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Table A1 (*continued*)

KATARI POLAO		Short–medium	Tapered	Smooth	Light translucent	Soft chalk belly
KATARIVOG		Medium	Tapered	Smooth	Semi-translucent	Very slender tip
LALAUSH		Short	Blunt	Rough	Brownish	Deep bran remnants
LALBINNI		Short	Blunt	Rough	Red-brown	Fully bran colored
LALBIROI		Medium	Tapered	Rough	Brown-white mixed	Partial bran + chalk
NAZIRSHAIL		Long slender	Sharp	Smooth	Semi-opaque	Popular aromatic type silhouette
PAIJAM		Medium	Blunt	Smooth	Semi-transparent	Roundish tip
REDCARGO		Short	Blunt	Rough	Deep red	Distinct cargo-rice trait
SHOMPA KATARI		Medium	Tapered	Smooth	Semi-translucent	Thin upper half
SHORNA5		Medium–long	Sharp	Smooth	Light translucent	Rounded belly
SUBOLLOTA		Medium-long	Tapered	Smooth	Semi-opaque	Slight curved spine
TURKISH BASMATI		Long slender	Very sharp	Smooth	Translucent-white	Very basmati-like needle grain

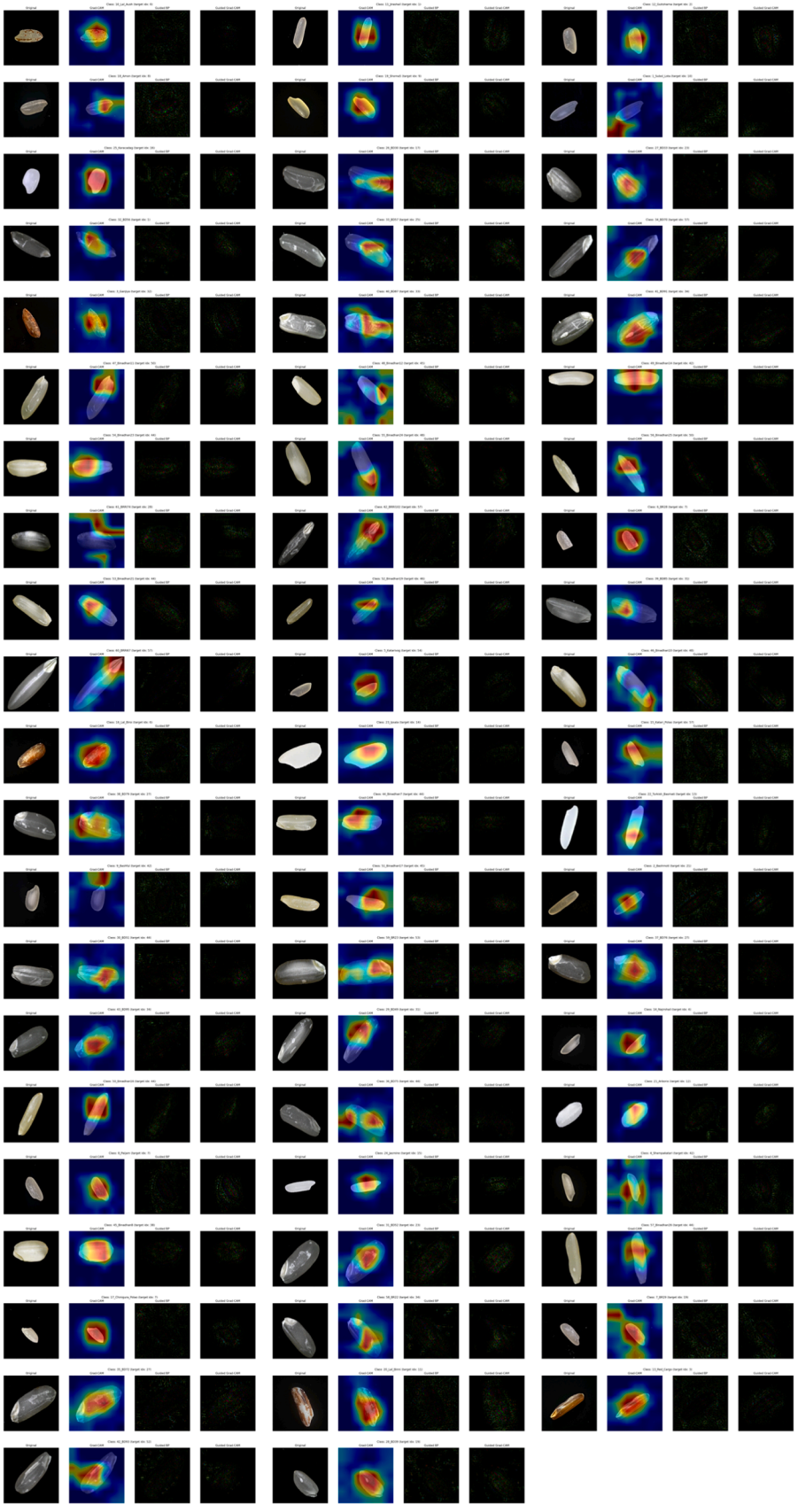


Fig. F1. GradCAM and Guided GradCAM of all Rice Kernels for Best CNN model (VGG16).

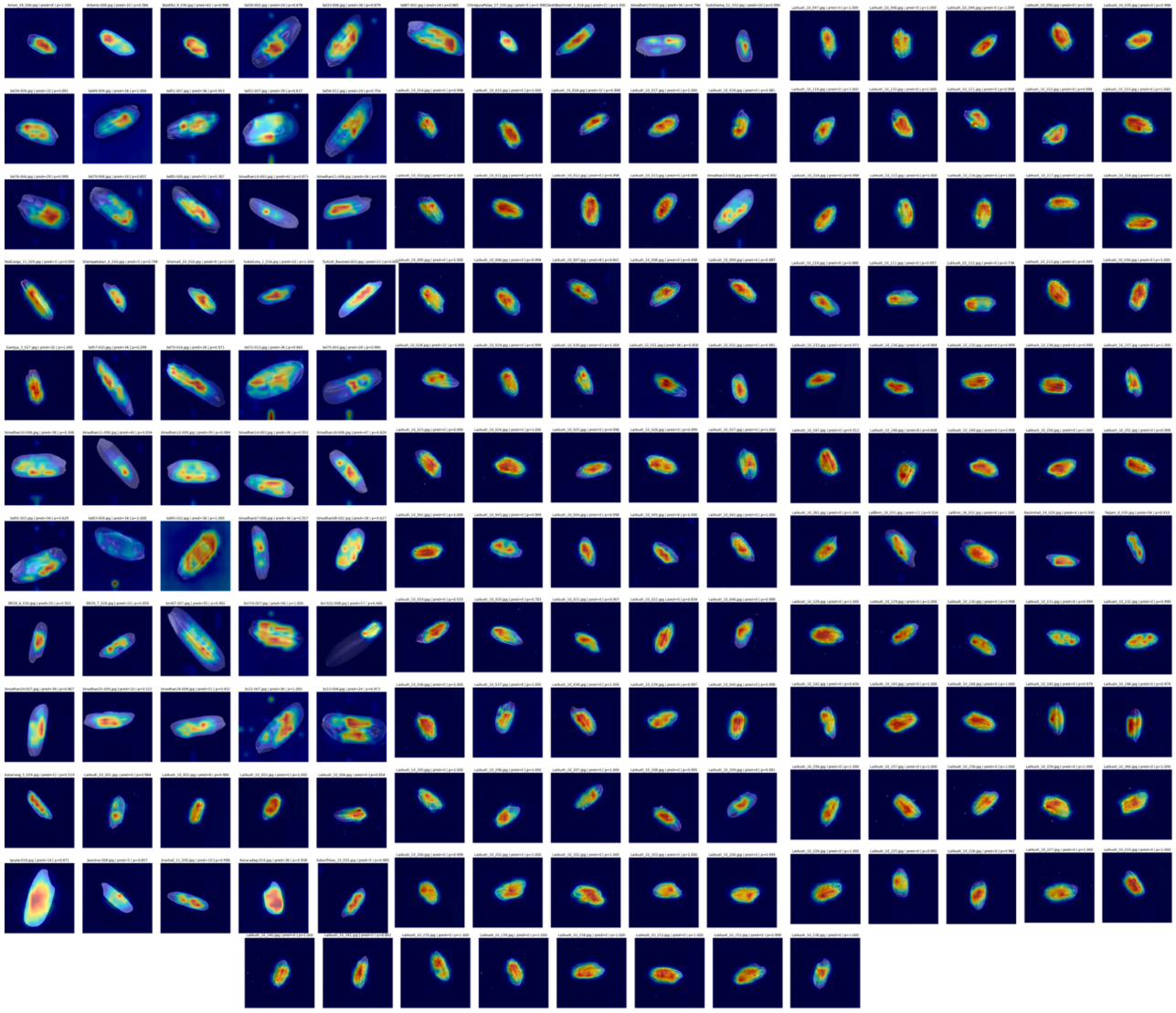


Fig. F2. Attention Map of all Rice Kernels for Best ViT model (Google's ViT B/16).

Data availability

Data will be made available on request.

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